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Business School

# The Impact of sentiment, credibility and negativity bias in online reviews on consumer trust and purchase intent in skincare industry

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## ABSTRACT

This study explores the combined impact of sentiment and credibility cues in online reviews on consumer trust and purchase intentions, with specific focus on skincare industry. Extending the elaboration likelihood model (ELM) the study explores on how consumers process review information through both central cue such as content quality, sentiment valence and peripheral cue such as verification and review helpfulness. A mixed method approach was use dand collected 307 responses through questionnaire capturing consumers reliance, trust in credibility signals and sensitivity to negative reviews and 2012 archived amazon reviews capturing behavioural insights from star ratings, helpfulness votes, review length and verified purchase tags. The study uses descriptive statistics, correlation analysis and multiple regression analysis with moderation and interaction to test hypothesis. The findings contribute theoretically by expanding on the eWOM research into highly personal industry like skincare and practically by providing insights on review credibility to brand managers.

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# CHAPTER I INTRODUCTION

## 1.1 Research Background

In recent years the growth of the internet and online technology has been rapid leading to people rely heavily on their day-to-day decision making. The increasing dominance of consumer generated content (CGC) has become an important factor of customer journey, shaping perceptions and purchase decisions. Online reviews play an important role in the ecommerce by providing social proof and reducing the confusion and information asymmetry between the buyers and sellers (Chevalier & Mayzlin, 2006). The rise in e commerce platforms such as Amazon, Tripadvisor and Trustpilot and their reviews are now seen as electronic word of mouth(eWOM) which gives a strong signal or influence consumers on their purchasing behaviors than traditional marketing communications (Zhu & Zhang, 2010). However, there is less attention given in understanding the type of reviews that are perceived as more credible and how credibility converts into consumer decision making across contexts such as in the skincare industry.

Prior research has identified that consumers use multiple cues to analyse and evaluate review credibility. Source based cues such as verified purchase tags gives signal to the consumers that the review is authentic, and the reviewer has genuinely bought and used the product. (Mudambi and Schuff ,2010) shows that verified reviews are perceived more helpful but, (Baek et al. 2012) shows that consumers only perceive them as credibility cues when they have conflicting opinions. The other crucial factor is the psychological influence of negativity bias while evaluating reviews under credible conditions. Psychology research shows that the negative information has stronger impact on decision making over positive information (Baumeister et al., 2001). (Herr et al., 1991; Park & Lee, 2009) also observed the effect of negative reviews were higher when buying high risk products. However, the negativity bias has not been explored sufficiently under the combination of credibility cues showing us a meaningful theoretical gap with practical implications for online retail and review platform design.

While both verification and detail have been studied individually most of the literature isolates these cues rather than studying the combining effects of these cues. This is a significant gap because consumers often rely on a single cue but multiple cues simultaneously when it comes to evaluating a review (Filieri, 2015). Furthermore, skincare consumers often spend significant time researching the products and the opinions of the reviewers that used the product (Mintel, 2022).

Taken the literature and the work done on the online reviews, the motivation for this research is to understand the identified gaps such as investigation of multiple credibility cues in combination and in isolation. Understanding how credibility is works in an underexplored domain of skincare industry where the opinion of the other user is critical for decision making. Identifying the presence of negativity bias among the consumers and the attributes that contribute to them. Finally, the methodological gap that combines survey data with behavioural review data. Addressing these gaps is important to advance both theoretical understanding of online review credibility and provide actionable insights for platforms designing review systems and marketing strategies in highly competitive consumer categories.

## 1.2 Research Aim, Questions and Objectives

The study's research question that is guiding is to understand *What is the combined impact of sentiment, credibility cues and negativity bias in online reviews on brand perception and consumer purchase intent in the skincare industry?* The Aim of this study is to examine how sentiment valence, credibility cues such as verified purchase labels, review length, star ratings, and helpfulness votes interact in influencing consumers trust in online reviews, their perception on the review and the factors the influencing purchase decisions by analyzing both attitudinal and behavioural factors the study seeks to bridge theoretical insights from consumer behaviour with practical implications for digital marketing and brand management in the skincare sector.

### 1.2.1 Research Questions

These are the following specific research question that this study aims to address:

1. To what extent does review sentiment (valence) influence consumer evaluations, and is there alignment between textual sentiment and star ratings in online reviews?

2. How do credibility cues such as verified purchase status and review reliance affect consumer trust in online reviews?

3. What role does review detail (length and elaboration) play in shaping consumer trust, and does purchase frequency moderate this relationship?

4. How does negativity bias manifest in consumer responses to online skincare reviews, and is its impact strengthened by credibility cues?

**1.2.2 Research Objectives**

1. To understand the relationship between review sentiment and consumer evaluations by analyzing whether positive reviews consistently correspond to higher product ratings, and whether negative reviews exert a stronger influence in line with the theory of negativity bias.

2. To examine the role of credibility cues in shaping consumer trust on signals such as verified purchase status, review reliance, and helpfulness votes, which acts as markers of authenticity and reliability.

3. To evaluate the effect of review detail and elaboration on consumer perceptions of credibility. Longer or more detailed reviews are often seen to improve trustworthiness, but how their impact needs systematic analysis, particularly when moderated by consumer purchase frequency.

4. To assess the combined effect between negativity bias and credibility signals. This includes determining whether negative reviews become more persuasive when combined with credibility indicators (e.g., a verified purchase label) compared to when such cues are absent.

5. To integrate insights from both consumer-reported perceptions and behavioural data. By combining evidence on attitudes (what consumers say they trust) with outcomes of actual review interactions.

## 1.3 Proposed Method

The methodological approach used in this dissertation uses both primary survey data and secondary review data to understand how sentiment and credibility cues in online reviews influence consumer trust, brand perception, and purchase intent. This study uses a multi-source methodological design, combining survey data with secondary data from Amazon. The rationale for this is to understand both attitudinal perception towards the credibility cues and behavioural response in practice. (Podsakoff, MacKenzie and Podsakoff, 2012) and (Venkatesh, Brown and Bala, 2013) explains the importance of the use of multiple data sources to improve external validity and avoid bias as single source approach can limit the generalizability.

The Survey method is well equipped to understand latent structures such as trust in verified reviews, trust in detailed reviews, review reliance, and sensitivity to negative reviews, which are not directly available or measurable through behavioural data. Survey is still considered a dominant tool in eWOM and consumer behaviour research to assess the psychological and perceptual factors (Ladhari and Michaud, 2015). In addition to the perceptual survey data this study uses archived Amazon data to leverage behavioural indicators such as helpfulness votes, star ratings, and review length. Online review datasets from major platforms such as Amazon are widely considered as rich and valid sources to understand consumer information processing (Luca, 2016). The use of the datasets has been prominent in the study of credibility cues, as amazon systematically records purchase verification and review helpfulness (Zhu and Zhang, 2010). So by combining primary and secondary datasets this study ensures a robust design that understands both self-reported attitudes and consumer behaviour. Analytically, this study uses regression analysis with interaction effects to test moderation hypothesis. This method is prominently used in consumer research to understand whether the relationship between the predictors and outcomes vary dependent on contextual factors (Hayes, 2017; Dawson, 2014). Models with interaction effect aligns with eWOM to understand the conditional effects rather than direct associations (Willemsen et al., 2011). In summary, this methodology uses survey and behavioural data and analysed through regression models with moderation to provide a theoretically and empirically sound way to explore how sentiment, credibility and bias in online review influence consumer.

### 1.4 Contributions

#### 1.4.1Theoretical contributions

1. (Filieri, 2016; Ladhari & Michaud, 2015) explored perception of credibility using only survey-based perception or secondary analysis of e-commerce data (Mudambi & Schuff, 2010). The study advances on the research on online consumer reviews, credibility, sentiment analysis by leveraging both self-reported survey data and behavioural data from the amazon reviews and bridges the gap.

2. This research seeks to advance on the theoretical understanding of credibility cues and sentiment in online reviews by placing them with broader framework of electronic work of mouth(eWOM) and consumer decision making. (Cheung et al., 2009) and (Filieri, 2016).

3. (Mudambi & Schuff, 2010) and (Willemsen et al., 2011) have shown credibility feature such as verified purchase badges, review length and star ratings contributes to influence perceived usefulness, this study builds up on the research to measure the effect of these credibility cues and compare with attitudinal data.

4. The study also extends the application of the Elaboration Likelihood Model (Petty & Cacioppo, 1986) to online review contexts. While prior research uses ELM framework to distinguish central cues such as message detail, quality and peripheral cues such as credibility and verification tags (Filieri & McLeay, 2014), This study uses both in a single framework to test their attitudinal and interactive influence by doing so expands the explanatory power of the ELM in digital context.

5. This study seeks to build on the work on the psychological effect of negativity bias (Baumeister et al., 2001) and seeks to explore the negativity bias present in online reviews that might influence the judgement and decision making of the consumer.

6. The study also seeks to refine (Filieri, 2015) study on self-reported perceptions and (Yin et al., 2014) platform level data research and theoretically demonstrates the how consumer attitudes

and real-world behaviour align or diverge when credibility cues and sentiment valence are present.

### 1.4.2 Practical Contributions

1. Using review reliance as a moderating factor, companies can use segmentation strategies to identify consumers that rely heavily on reviews for purchase and target them with credible review formats.

2. As the study checks the importance of credibility cues and how much it influences consumer decision, the findings can be leverages by platforms or brands to encourage verified purchases and provide incentives for longer detailed reviews to enhance brand reputation.

3. The study tests the negativity bias under credible condition; the findings can be used to understand the impact of a credible negative review and target the review with swift response to avoid any reputational damages.

4. The findings from the behavioural data can provide insights to review platforms on reviews that are perceived credible and design review systems that shows the perceived credible reviews first guiding consumers towards trustworthy information.

5. Given skincare has higher consumer involvement and reliance on peer reviews, brands can adapt their digital communication strategies by promoting credible reviews in product pages, influencer collaborations and social commerce platforms. Since the study uses attitudinal data and behavioural data, managers can take confidence policy changes for practice.

## CHAPTER II LITERATURE REVIEW

### 2.1 Introduction

This chapter showcases a detailed review of the theories and concepts which are relevant to the impact of sentiment, credibility, and negative bias in online review on consumer perception of skincare brands among consumers. It defines core constructs, such as online review sentiment, credibility indicators, and bias, along with an overview of the elaboration likelihood model (ELM). The first section explains the dependent, independent and mediating variables with related theory. The chapter then develops by examining how consumers interpret online reviews in the skincare context, followed by analyses of sentiment valence and negativity bias, credibility and content quality, ELM theory, and the effect of reviews on brand trust and purchase intent. It concludes by identifying research gaps and presenting hypotheses and research model.

### 2.2 Definition and Theory of Related Concepts

#### 2.2.1 Online Reviews and Consumer Perception

Online reviews function as a form of electronic word-of-mouth (eWOM), which has been widely recognized as one of the most powerful drivers of consumer decision-making in digital marketplaces (Chevalier & Mayzlin, 2006; Filieri, 2015). Unlike traditional advertising, reviews are perceived as more authentic and unbiased because they are from fellow consumers rather than firms. Research consistently shows that consumers increasingly substitute eWOM for interpersonal recommendations, especially where product performance cannot be known before purchase (King, Racherla & Bush, 2014).

In the absence of physical trials, skincare shoppers heavily rely on reviews as social proof to infer product effectiveness and safety (Prashant Parmar, Upadhyay and Sharma, n.d.). Survey evidence shows that 87% of consumers rely on online reviews before choosing a local business, and nearly 79% trust those reviews as much as personal recommendations from friends or family (Qahri-Saremi and Montazemi, 2022). Survey evidence also indicates that approximately 82% of consumers read product reviews before making purchase decisions, and 93% say that online reviews influence their choices (Chen et al., 2022). An industry analysis identified that items with at least five reviews have a 270% higher conversion rate compared to those with no reviews (Pooja and Upadhyaya, 2022). These combined results show how critical and important the role of review is in shaping consumer perception and behavior.

### 2.2.2 Consumer Perception in the Skincare Industry

Skincare brands and industries are prone to intense criticism and scrutiny over reviews from the consumers because of the personal nature of these products. Consumers who are using the product look for other's experience and information about the quality of the scent, texture or any abnormal reactions caused. Positive online customer reviews can serve as a persuasive tool. They increase the perceived value of unknown products and help builds trust, otherwise be difficult to establish (Sung, Chung and Lee, 2023). The same paper also discusses how positive reviews have increased brand image and credibility in the market. However, negative reviews can damage a brand's reputation and deter potential buyers; negative word-of-mouth has been linked to significantly more negative brand perceptions and reduced purchase intentions in skincare and beauty contexts (Dr. Satish Uplaonkar and Dr. Sharangoud Biradar, 2019). Showing that trust in critical for experience-based purchases like skincare, where product performance is unknown until it's being used. Online reviews and ratings play an important role in reducing the perceived risk and increasing confidence in purchase decisions. In summary, online reviews in the skincare sector are a double-edged sword that can greatly improve or affect consumer perceptions of a brand.

### 2.2.3 Sentiment Valence and Negativity Bias in Online Reviews

Sentiment valence refers to the overall positive or negative tone of a review (often reflected in the review's language or its star rating). (Ho-Dac, Carson and Moore, 2013) found that an accumulation of positive online customer reviews increased trust in lesser-known brands, improving their market performance. IN contrast to that even a few negative reviews can review can raise red flags: (Bae and Lee, 2011) observed that consumers seeking information on a well-established product still deemed a negative review (especially if posted on an independent community forum) as highly credible and influential in their perceptions. (Purnawirawan, De Pelsmacker and Dens, 2012) demonstrate that positive reviews improve brand attitudes and intentions, while negative reviews lead to skepticism and reduced purchase intent. Additionally, consumers are about 4 times more likely to choose a highly rated product over a low rated one in categories like skincare, reflecting the sentiment heuristics on decision making (Duan, Chen and Huo, 2019).

However, the influence of sentiment is not simply symmetric between positive and negative information. Negative information has a greater effect on perceptions than comparably extreme positive information (Wu, 2013) (Sen and Lerman, 2007). In online reviews, the negativity bias implies that the one-star review will tend to have more than a five-star review on consumer judgement. (Skowronski and Carlston, 1989) paper on information processing shows that people give disproportionate attention and weight to negative signals, which are perceived as more diagnostic or indicative of true quality problems (Colmekcioglu et al., 2022). Applying this to eWOM, (Sen and Lerman ,2007) found clear evidence of negativity bias: consumers found negative reviews more informative and credible than positive reviews for utilitarian products, leading those negatives to be more influential on purchase decisions. Whereas more experienced and analytical minded customers might scrutinize content quality, but they are not immune to a well detailed negative review (Qahri-Saremi and Montazemi, 2022). (Guo, Wang and Wu, 2020) shows that the perceived review credibility significantly amplified the effect of negative eWOM on consumer decisions, suggesting that a negative review that readers trust can be damaging. In conclusion, negative bias generally prevails in online reviews: consumers tend to remember and rely on negatives more than positives when forming impressions of a brand (Rozin and Royzman, 2001).

### 2.2.4 Review Credibility and Content Quality

Review credibility refers to the extent to which consumers perceive online review as believable, truthful and reliable. Literature consistently finds that credible reviews hold greater influence on consumer attitudes and decisions than less credible ones, acting as a central persuasive force in the presence of sufficient consumer engagement (Jiménez and Mendoza, 2013).

It is identified that numerous factors such as argument quality, clarity, depth and logic of the review content are key to the review’s credibility. (Cheung and Thadani, 2012) applied ELM and found that text quality, a central cue, was the primary factor affecting review credibility. The study also mentions that short or vague reviews are often perceived as less credible, partly due to their low informational value. Cheung et al. also noted that readers use peripheral credibility cues such as source credibility, review consistency and review sidedness (pros and cons) when evaluating credibility. Moreover, the consistency among multiple reviews acts as a marker for credibility. If many reviewers report the same pros and cons about a product, each individual review's credibility is improved (Reyes-Menendez, Saura and Martinez-Navalon, 2019). Additionally, a single high positive review among many negative reviews is viewed with scepticism and is considered inauthentic by the consumers (Mayzlin, Dover and Chevalier, 2014). Platforms which often provide meta judgements like helpfulness vote or count also influence the perceived credibility, as a highly rated review signals that other readers vetted its value, indirectly enhancing its credibility (Chen et al., 2016). If the review is suspected to be "too good to be true" consumers may ignore it or develop negative sentiment towards the brand. The way these credibility factors interact with basic sentiment cues can be understood using the Elaboration Likelihood Model, as explained in the next section.

**2.2.5 Consumer reliance on reviews**

Consumer reliance on online reviews has become an important factor of decision-making in digital commerce, as reviews are seen to reduce uncertainty and information asymmetry between buyers and sellers. Research consistently shows that consumers actively consult reviews before making purchases, especially for high-involvement or experience-oriented products such as skincare (Park & Lee, 2009; Filieri, 2015). Reliance on reviews is not uniform; it varies by demographics, such as age and purchasing experience. Younger consumers, who are digitally native, demonstrate stronger tendencies to depend on peer-generated reviews as opposed to traditional advertising (Zhu et al., 2014).

Reliance is also shaped by perceived credibility of reviews. When consumers trust that reviews are authentic and unbiased, they are more likely to rely on them in purchase decisions (Filieri & McLeay, 2014). Moreover, frequent online purchasers develop habitual reliance on reviews, using them as decision heuristics (Mudambi & Schuff, 2010). Studies further indicate that reliance is moderated by product type: search goods, where quality can be assessed prior to purchase which may require less review reliance, while experience goods like skincare increase dependence on peer feedback (Hu et al., 2009).

Reliance is not only deals with frequency but also depth of engagement. Customers read multiple reviews in detail and some skim other cues such as average rating or helpfulness votes. (Zhang et al., 2014). This differentiation links closely with persuasion models such as the Elaboration Likelihood Model (Petty & Cacioppo, 1986), where reliance reflects either central or peripheral processing routes. Overall, consumer reliance is a construct shaped by trust, demographics, and product category, making it a crucial variable in understanding credibility and purchase behaviour in online review contexts.

### 2.2.6 Review helpfulness as a social proof signal

The concept of review helpfulness is central to understanding how consumers evaluate and adopt online information. Helpfulness votes, usually provided by readers as signals of whether a review was useful and acts as a form of social proof (Cialdini, 2009). A review with more helpfulness from others is perceived as more credible and influential, regardless of its content quality (Mudambi & Schuff, 2010). Multiple studies confirm that helpfulness is positively associated with review length and richness of detail, as longer reviews tend to be seen as more informative and diagnostic (Korfiatis et al., 2012; Yin et al., 2014). However, excessively long or overly technical reviews may reduce helpfulness perceptions if they overwhelm readers (Forman et al., 2008). Helpfulness is also shaped by valence; negative reviews, especially when detailed, often receive more helpfulness votes than positive reviews due to negativity bias (Baumeister et al., 2001; Lee, 2009).

Verified purchase status further boosts helpfulness, as consumers find reviews from confirmed buyers more authentic and trustworthy (Filieri, 2016). Additionally, review helpfulness functions as a mediator in the relationship between review content and consumer behaviour, since it determines which reviews are most visible and influential in digital marketplaces (Fang et al., 2016). In sum, helpfulness votes influences consumer judgment and provide critical insights into how social validation shapes the impact of eWOM.

### 2.2.7 Elaboration Likelihood Model (ELM)

The Elaboration Likelihood Model (ELM) proposed by Petty and Cacioppo (1986) remains one of the most influential frameworks in persuasion research and has been widely applied to online consumer behaviour. The model posits two distinct routes, the central route and the peripheral route. Central route involves careful and thoughtful consideration of the message where peripheral route consumers rely on simple cues such as source credibility, popularity signals and other heuristics. The ELM provides a theoretical structure for understating how consumers process online review (Kitchen et al., 2014).

With relevance to online review, detailed arguments and sentiment content represent central route processing as consumers engage in more thoughtful elaboration when reviews have rich information (Cheung et al., 2009). However, cues such as verification badge or number of helpfulness votes acts as peripheral cues influencing customer decision without needing extensive thought or effort (Filieri & McLeay, 2014; Sher & Lee, 2009). A highly involved consumer more likely to process reviews via central route, whereas less involved consumers depend on peripheral cues (Chang & Wu, 2014).

Recent literature further shows that ELM provides a robust framework for combining credibility and sentiment research in online environments. (Filieri ,2016) shows that consumers assess both the quality of review arguments and source credibility signals side by side, aligning with ELM dual route approach. This makes ELM particularly relevant for understanding how sentiment, credibility, and social proof interact to shape purchase intent in online marketplaces.

### 2.3 Gap Identification

While there is multiple research that focuses on electronic word of mouth and online reviews, significant gaps still remain on how credibility cues and sentiment interact to form consumer perceptions. The previous studies have shown the importance on online reviews in influencing consumers and their purchase decision (Chevalier & Mayzlin, 2006; Mudambi & Schuff, 2010). Still most of the work is on credibility and sentiment independently, rather than in combination. Research on sources credibility has largely focused on structural indicators such as verification status or reviewer reputation (Filieri, 2016), while studies on sentiment have shown the importance of the role of review valence in shaping perceptions (Baumeister et al., 2001; Yin et al., 2014). However, there is limited study on how these two streams intertwine, specifically whether source credibility cue and message cue increase or weakens the effect of sentiment on consumer trust and helpfulness.

Moreover, most of the existing literature relies on single platform analysis rather than integrating survey-based psychological attributes with observational review data. This creates a gap in understanding both the perceptual (survey-based trust and reliance) and behavioural (helpfulness votes, review structure) dimensions of credibility. Additionally, while the negativity bias is well established in psychology (Baumeister et al., 2001), few studies have systematically tested how it interacts with credibility cues in digital review contexts. Addressing these gaps is important because it allows for a clearer understanding of how consumers weigh reviews not only by what is said (sentiment/detail) but also by who says it and under what conditions (verification, helpfulness signals, etc.).

### 2.4 Hypothesis Development

*H1: There is a positive relationship between the sentiment valence of review text and the star rating assigned.*

Sentiment valence in textual reviews is often assumed to align with quantitative star ratings, yet this assumption is not always empirically validated. Research shows that reviews with more positive wording typically correspond to higher star ratings, whereas negative textual sentiment aligns with lower ratings (Zhang et al., 2014; Fang et al., 2016). However, discrepancies can occur, such as when consumers assign moderate ratings despite strongly negative textual content (Yin et al., 2014). When reviews show more positive sentiment or language they are usually accompanied with higher ratings, showing the consistency between verbal and quantitative signals and negative textual sentiment to have lower ratings (Zhang et al., 2014). In context of skincare, it is particularly relevant since the experimental cues such as texture, fragrance is highly subjective and often emotionally heavy(Sung, Chung, & Lee, 2023). If the textual sentiment and ratings are different, it can create confusion to the consumer and weakens their perceived review credibility (Filieri, 2015). Therefore, understanding this relationship is important because it validates whether automated sentiment measures capture consumer perceptions in ways consistent with explicit rating behaviour. Therefore, this study hypothesizes that positive sentiment in reviews will be positively associated with higher star ratings.

*H2: Consumers who rely more on online reviews are more likely to trust verified reviews, and verified reviews receive significantly more helpfulness votes than unverified reviews.*

The second hypothesis explores whether verified purchase status enhances perceptions of review helpfulness, with star rating as a potential moderator. A important factor in online review is, whether the reviewer can be verified as a genuine purchaser, since verified purchase status serves as a source credibility cue that reduces uncertainty for potential buyers (Zhu & Zhang, 2010). Reviews without verification may be seen as fabricated, incentivized, or biased, particularly in industries where fake reviews are prevalent (Luca & Zervas, 2016). The Elaboration Likelihood Model (ELM) mentions that peripheral cues such as verification can increase persuasion when consumers rely on heuristic shortcuts (Petty & Cacioppo, 1986). This is particularly relevant in the skincare industry, where consumers face heightened risk due to concerns over product quality, side effects, and effectiveness (Chen et al., 2022). Prior studies demonstrate that reviews with verification badges tend to attract more helpfulness votes compared to unverified reviews (Forman, Ghose, & Wiesenfeld, 2008). Beyond verification, helpfulness votes act as a form of social proof, indicating the collective endorsement of certain reviews by the wider consumer community (Filieri, 2015). Consumers may view a verified five-star review as particularly persuasive, while a verified one-star review may appear more damaging, since the verification makes the negative evaluation more believable (Mudambi & Schuff, 2010).

*H3: Consumers place greater trust in detailed reviews, and detailed review, especially when verified are more likely to be rated as helpful.*

The third hypothesis explores the relationship between review length and perception of review credibility or helpfulness in amazon dataset. Longer reviews typically contain richer information and signals consumers of greater reviewer effort, improving the perceived credibility (Pan & Zhang, 2011). The cues provide further information for understanding which reduces uncertainty and increase confidence in purchase decisions, especially in skincare (Mudambi & Schuff, 2010). However, literature also warns of a threshold effect, where reviews that are excessively long may reduce readability and overwhelm consumers (Korfiatis, García-Bariocanal, & Sánchez-Alonso, 2012) excessively long reviews can risk information overload, reducing readability and effectiveness (Baek, Ahn, & Choi, 2012). The studies show curvilinear effect in some situations where moderately long reviews are considered most useful. This builds on diagnosticity theory (Herr, Kardes, & Kim, 1991), suggesting that information perceived as more diagnostic has a stronger influence on judgments. In skincare, where product results vary individually, consumers are likely to value detailed reviews that provide situational context, thereby making length a strong predictor of credibility perceptions. By testing the role of review length, H3 contributes to understanding whether longer, more detailed reviews consistently enhance perceptions of credibility and helpfulness in the skincare sector.

*H4: Negative reviews receive more helpfulness votes than positive reviews, and this effect is amplified when reviews are both detailed and verified*

The fourth Hypothesis explores whether negative reviews have more weight and have stronger effect over than positive reviews, reflecting psychological phenomenon of negativity bias. Negativity bias theory (Baumeister et al., 2001) posits that negative information has a greater impact on judgment and decision-making than positive information of equal intensity. In online reviews the negative reviews have been shown to disproportionately affect brand and make consumers avoid purchases (Lee, Park, & Han, 2008). This effect is particularly salient in the skincare industry, where concerns over side effects or product inefficacy heighten sensitivity to negative content (Dr. Satish Uplaonkar & Biradar, 2019). A single detailed review can overshadow multiple positive reviews.

Moreover, negativity may interact with credibility cues. When a negative review are verified and detailed the impact of it is amplified because they are seen as more trustworthy and diagnostic (Filieri, 2016). Attribution theory also supports that consumers treat negative experiences as stronger evidence of product performance over positive ones, since dissatisfaction is harder to fake (Mizerski, 1982). Prior research shows that consumer engage more with negative reviews by spending more time and comparing them to positive review (Rozin & Royzman, 2001). Consumers may rely on negative reviews to avoid potential harm, thus providing them an unequal weight when it comes to decision making. H4 therefore tests negativity bias particulary when combined with credibility cues in consumer perceptions, extending negativity bias theory into online review context for skincare

# CHAPTER III METHODOLOGY

## 3.1 Research Design

This study uses a positivist research philosophy, where it focuses on quantifiable measurement, hypothesis testing and the search for generalizable patterns through statistical modeling (Saunders et al., 2019). By using constructs such as trust, credibility and review helpfulness into measurement indicators, this study aligns with positivist theory of explaining causal relationships through objective data analysis. This study uses a quantitative, explanatory, cross-sectional research design to understand how online review credibility cues and consumer reliance on reviews affect brand perception and purchase intent. A quantitative design is the most appropriate as it allows for statistical modelling of causal relationship between variables and facilitating the hypothesis (Creswell & Creswell, 2018).

The design integrates two complementary datasets; The first dataset is a structured survey that captures consumers' self -reported attitudes and perceptions and the secondary dataset consisting of Amazon skincare reviews capturing observable behaviour such as helpfulness votes. Combining these approaches supports methodological triangulation, which improves the validity and avoids any potential common method bias (Venkatesh, Brown & Bala, 2013). Regression and moderation analyses forms the main analytical strategy for this study, these models are commonly use in consumer research to understand how one variable influences the other variable under specific contextual conditions (Hayes, 2018).For example, The survey models tests whether consumer reliance on reviews increases in their trust in verified reviews or detailed reviews, while amazon models assess whether review length boosts the impact of negativity on perceived helpfulness. By using both the datasets this design explores both subjective and consumer beliefs and objective marketplace dynamics, providing a rich and robust understanding of credibility in online skincare review.

## 3.2 Data sources and sampling

### 3.2.1 Primary data - *Survey*

Primary data were collected using an online survey distributed through Microsoft Forms. A total of 307 valid responses were obtained from skincare consumers in the UK. The survey was disseminated via convenience and snowball sampling across university networks and social media platforms. While non-probabilistic, such approaches are common in exploratory consumer behaviour research where access to large-scale random samples may be impractical (Etikan, Musa & Alkassim, 2016).

The survey instrument was structured into three parts:

1. **Credibility cues:** Trust in verified reviews, trust in detailed reviews, and sensitivity to negative reviews.

2. **Consumer reliance:** Extent of dependence on online reviews for purchase decisions.

3. **Demographic and behavioural controls:** Age (continuous) and purchase frequency (categorical numeric).

Responses were measured using 5-point Likert scales (1 = strongly disagree, 5 = strongly agree), which are widely adopted for measuring perceptions and attitudes in consumer research (Joshi et al., 2021). Purchase frequency was coded as a numeric variable (e.g., 1 = weekly, 2 = monthly, 3 = every 2–3 months, 4 = rarely). The achieved sample size of 307 is consistent with statistical requirements for multiple regression, where 10–15 cases per predictor variable are typically recommended (Hair et al., 2019). However, it is less in scale , the sample provides sufficient power for the intended analyses.

### 3.2.2 Secondary data - *Amazon reviews*

Secondary data were sourced from 2,012 consumer reviews of skincare products on Amazon, compiled by the McAuley Lab (University of California, San Diego) this dataset contains anonymized user-generated reviews for skincare products posted between 1996 and 2023. Amazon was selected because it is a leading global e-commerce platform where review credibility cues such as verified purchase and helpfulness votes are prominently displayed, making it ideal for studying consumer trust signals (Chevalier & Mayzlin, 2006; Zhu & Zhang, 2010).

The dataset contained:

1. **Review text:** Used for sentiment and length analysis.

2. **Star rating:** 1–5 scale, reflecting consumer product evaluations.

3. **Helpfulness votes:** Count of “helpful” marks from other consumers (dependent variable).

4. **Verified purchase status:** Binary indicator (0 = unverified, 1 = verified).

Data preprocessing involved removing duplicates, irrelevant items (non-skincare), and extreme outliers (e.g., unusually long reviews). Helpfulness votes were log(x+1) transformed to correct skewness, a standard approach in review helpfulness research (Yin, Bond & Zhang, 2014). Sentiment analysis was applied to review text using DistilBERT, a transformer-based natural language model that captures contextual sentiment more accurately than lexicon-based methods such as VADER (Devlin et al., 2019). Scores ranged from –1 (very negative) to +1 (very positive), which were inverted in negativity bias tests to construct a negativity index. Review length was changed as the word count of each review, reflecting message detail. Prior work indicates that longer reviews provide diagnostic value and are perceived as more credible (Mudambi & Schuff, 2010).

### 3.3 Variables and Measurements

#### 3.3.1 Survey Variables

**Dependent variables:**

**Trust in verified reviews** – consumer confidence in reviews with a verification badge.

**Trust in detailed reviews** – perceived credibility of longer, information-rich reviews.

**Sensitivity to negative reviews** – degree of influence from negative content.

**Independent variable:**

**Review reliance** – extent to which consumers depend on reviews when purchasing skincare products.

**Moderators and controls:**

**Purchase frequency** – frequency of buying skincare, moderating the relationship between reliance and trust.

**Age** – continuous control variable to account for demographic influences.

These constructs are grounded in studies showing that credibility cues and consumer reliance interact to shape trust and purchase intent (Filieri, 2016; Qahri-Saremi & Montazemi, 2022).

### 3.3.2 Amazon variables

**Dependent variable:**

**Helpfulness votes** – log-transformed count of helpful votes as an outcome of credibility.

**Independent variables:**

**Verified purchase** – presence of verification badge (source credibility).

**Sentiment valence** – polarity of review text (–1 = negative, +1 = positive).

**Review length** – word count of review text (message credibility).

**Moderators and controls:**

**Star rating** – included as a control, reflecting overall product satisfaction.

**Interaction terms** – e.g., Verified × Rating, Negativity × Length × Verification, used to test moderation effects.

These variables align with eWOM literature that positions helpfulness as a form of social proof, influenced by both source and content cues (Mudambi & Schuff, 2010; Filieri, 2015).

### 3.4 Data Analysis

The study follows a structured, multi- step process designed to evaluate and ensure validity of the results, with descriptive statistics, followed by correlation analysis, regression, moderation testing, and the use of interaction plots to understand complex effects. The approach is considered as best practice in empirical consumer behaviour where quantitative analysis provides solid basis for testing hypothesis and identifying relationships (Hair et al., 2019). First step involves descriptive statistical analysis to measure mean, median and shape of distribution helping us to understand the normality issues that may cause statistical problems (Tabachnick & Fidell, 2019). Normalizing skewed data in this way ensures that regression assumptions, particularly regarding error distribution, are not violated (Kline, 2015). Pearson correlation analysis on both datasets to construct validity and issue of multicollinearity. Correlation analysis provides preliminary insights whether the hypothesized relationship is feasible or not before regression testing (Cohen et al., 2014).

Ordinary least squares (OLS) a central analytical method used in the study for testing directional hypothesis and their effect on independent variables on dependent variables with controlling for moderators and covariates (Wooldridge, 2015). Predictor s and moderators with mean centered prior to create interaction terns, reducing multi collinearity and improving interpretability (Hayes, 2018). Finally, interaction plots to illustrate the relationship between an independent and dependent variable depending on the change in moderator (Aiken & West, 1991). The analytical structure aligns with methods in eWOM research, which often leverages regression, moderation and visualization techniques to explore the hypothesis (Filieri, 2016; Ismagilova et al., 2020).

### 3.5 Ethical considerations

This study had various ethical considerations during the design given it’s significance and importance. Firstly, prior to the data collection a formal approval form was obtained from the university’s ethics committee. Along with the ethics application, the survey questionnaire was checked and got its approval from the supervisor prior to collection. For the primary data, the survey respondents were informed about the study in detail and provided an consent before participation. They participant were assured anonymity with no personal information was collected from them. All data collected was stored securely in university cloud account that is password protected and compliant with GDPR to protect participant privacy. The secondary data with amazon dataset was publically available data and all personal identifier was removed from the dataset during data cleaning. This approach was aligning with ethical best practices for research using digitally traceable data (Townsend & Wallace, 2016).

This study documented all data cleaning procedures and steps taken to withhold the ethical standards to protect participant from any data leakage related to their personal information. The data collected for the purpose of the study will be destroyed after the study to ensure confidentiality of participants.

### 3.6 Limitations

While this study has a robust methodological structure, there are certain limitations that needed to be addressed. The survey dataset used was sufficient for running regression but as limited this reduces the generalizability of the findings across wider consumer groups and demographics. The secondary dataset after filtering, removing duplicates, irrelevant entries for the required consistency can reduce the diversity of consumer perspective. Due to the survey being sent through university channels and social media the self- reported survey may attract like-minded people, causing unconsciously bias in sampling. The Archived amazon dataset did not have other important factors such as reviewer expertise, platform-specific features, or multimedia elements narrowing the insights of how consumers assess credibility through the mentioned cues.

## CHAPTER IV FINDINGS

This chapter displays the result of the statistical analyses, which are categories into four sections. Section 4.1 covers the descriptive analysis of sample distribution; Section 4.2 presents the descriptive statistical analysis of key variables of both survey and Amazon. Section 4.3 provides correlation analysis of both survey and Amazon dataset, finally Section 4.4 then reports the results of the regression analyses across survey and review data to test the hypotheses, organized under four sub-sections.

### 4.1 Descriptive analysis of sample distribution - *(Survey)*

| Classification | Count (Frequency) | Percentage |
|---|---|---|
| **Age Group** | | |
| ***18 - 24*** | 154 | 50.16% |
| ***25 - 34*** | 119 | 38.76% |
| ***35 - 44*** | 21 | 6.84% |
| ***45+*** | 13 | 4.23% |
| | | |
| **Purchase Frequency** | | |
| ***Monthly*** | 110 | 35.83% |
| ***Weekly*** | 92 | 29.97% |
| ***Every 2 - 3 months*** | 71 | 23.13% |
| ***Rarely*** | 34 | 11.07% |

Table 1: Descriptive Analysis of Sample Distribution

Table 1 presents the descriptive analysis of the primary survey sample distribution based on respondents’ age group and purchase frequency of skincare products, the majority of respondents (50.16%) belong to the age group 18- 24, followed by 25- 34 years (38.76%) and approximately 11% of respondents older than 35 and above. This suggests that the sample is predominately composed of youngest consumers, who are generally more active in online product research and review engagement for skincare products.

In terms of purchase frequency, most of the respondents (35.83%) reported purchasing skincare products monthly. followed by weekly purchases (29.97%). Around 23.13% purchase products every 2-3 months, while 11.07% purchase them rarely.

### 4.2 Descriptive analysis of Primary (Survey) and Secondary data (Amazon reviews)

The table below presents the descriptive statistics of the survey response (panel A) measuring respondents' perception and behavior regarding online skincare product reviews and of the amazon reviews (panel B) measuring review variables such as helpfulness, rating, verified status, length and their sentiment.

**Panel A (Survey)**

| Variable | N | Min | 25% | 50% | 75% | Max | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|---|
| *Review dependence* | 307 | 1 | 4 | 4 | 5 | 5 | -1.109 | 4.212 |
| *Trust in detailed reviews* | 307 | 1 | 4 | 4 | 5 | 5 | -0.733 | 3.02 |
| *Trust in verified reviews* | 307 | 1 | 4 | 4 | 5 | 5 | -0.544 | 2.692 |
| *Trust in verified and detailed.* | 307 | 1 | 3 | 4 | 4 | 5 | -0.909 | 5.069 |
| *sensitivity to negative reviews* | 307 | 1 | 3 | 4 | 4 | 5 | -0.717 | 3.388 |
| *Age* | 307 | 21 | 21 | 21 | 29.5 | 47.5 | 1.296 | 4.333 |
| *Purchase frequency* | 307 | 1 | 1 | 2 | 3 | 4 | 0.417 | 2.182 |

**Panel B (Amazon reviews)**

| Variable | N | Min | 25% | 50% | 75% | Max | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|---|
| *Review helpfulness* | 2012 | 0 | 0 | 0 | 0 | 133 | 16.697 | 366.339 |
| *review rating* | 2012 | 1 | 4 | 5 | 5 | 5 | -1.438 | 4.029 |
| *Verified purchase status* | 2012 | 0 | 0 | 1 | 1 | 1 | -0.054 | 1.003 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| ***Review length (words)*** | 2012 | 1 | 19 | 47 | 96.25 | 1511 | 4.37 | 40.472 |
| ***Review sentiment score*** | 2012 | -1 | -0.718 | 0.989 | 0.999 | 1 | -0.908 | 1.937 |

*Table 2: Descriptive statistics of the survey (Panel A) and Amazon review (Panel B) datasets*

In Panel A**,** trust-related variables such as *Review dependence*, *Trust in detailed reviews*, *Trust in verified reviews,* and *Trust in verified + detailed reviews* all have relatively high mean scores (M ≈ 4.0–4.3 on a 5-point Likert scale) with low standard deviations (SD ≈ 0.77–0.94), indicating generally high and consistent trust among respondents. *Sensitivity to negative reviews* also shows a moderately high mean (M = 3.79), while *Purchase frequency* has a lower mean (M = 2.14) on a 4-point scale, suggesting average exposure to online reviews. *Age* ranges between 21 and 47.5 years (M = 26.68, SD = 7.00). All survey variables report skewness and kurtosis values within acceptable limits (|skew| < 2, |kurtosis| < 7), meeting the general assumptions of normality for parametric testing (Mohamed Elarbi Boumidouna, 2024).

In Panel B**,** the distribution patterns are more varied. *Review helpfulness* (measured by helpful votes) shows extreme positive skew (skewness = 16.697) and leptokurtosis (kurtosis = 366.339), which is consistent with prior research showing that most online reviews receive no or few helpful votes, while a small number receive disproportionately high counts (Chevalier & Mayzlin, 2006; Yin et al., 2014). Review length (words*)* also exhibits high skewness (4.37) and kurtosis (40.47), indicating that while most reviews are short, a few are extremely long. In line with established practice in electronic word-of-mouth (eWOM) research (e.g., Mudambi & Schuff, 2010; Baek et al., 2012), both variables were log(x + 1) transformed to reduce skewness and better meet regression assumptions. *Review rating* is negatively skewed (skewness = –1.438), reflecting a concentration of high-star reviews (4–5 stars), and *Verified purchase status* is binary with a near-equal split (M = 0.513), showing a balanced distribution between verified and unverified reviews. *Review sentiment score* was normalized between –1 and +1 and shows acceptable skewness and kurtosis.

### 4.3 Correlation Analysis

Correlation analysis is a mathematical or statistical tool that is used to understand how one variable affects another and their degree of closeness. The correlation coefficient measures the linear relationship between variables to test the correlation is statistically significant and not random (JMP Statistical Discovery, 2025). To understand the correlation between these variables or statements, the Pearson's correlation analysis of respective variable a was conducted using python. The below table shows the correlation for both survey (panel A) and Amazon review (panel B).

| **Panel A - Survey** | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | **Mean** | **SD** | **1** | **2** | **3** | **4** | **5** | **6** | **7** |
| 1 | ***Review dependence*** | 4.293 | 0.828 | - | | | | | | |
| 2 | ***Trust in detailed reviews*** | 4.143 | 0.866 | 0.365 | - | | | | | |
| 3 | ***Trust in verified reviews*** | 4.186 | 0.776 | 0.444 | 0.364 | - | | | | |
| 4 | ***Trust in verified and detailed.*** | 3.932 | 0.791 | 0.325 | 0.20 | 0.276 | - | | | |
| 5 | ***sensitivity to negative reviews*** | 3.798 | 0.942 | 0.466 | 0.204 | 0.378 | 0.341 | - | | |
| 6 | ***Age*** | 26.68 | 7.002 | -0.130 | -0.034 | -0.084 | -0.212 | -0.293 | - | |
| 7 | ***Purchase frequency*** | 2.14 | 0.965 | -0.191 | -0.091 | -0.122 | -0.236 | -0.231 | 0.334 | - |

| **Panel B - Amazon review** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | **Mean** | **SD** | **1** | **2** | **3** | **4** | **5** |
| 1 | ***Review helpfulness*** | 1.003 | 5.185 | - | | | | |
| 2 | ***Review sentiment score*** | 0.411 | 0.848 | -0.024 | - | | | |
| 3 | ***Verified purchase status*** | 0.513 | 0.5 | 0.067 | -0.076 | - | | |
| 4 | ***Review length (words)*** | 75.412 | 95.719 | 0.084 | -0.053 | -0.413 | - | |
| 5 | ***review rating*** | 4.188 | 1.19 | -0.015 | 0.659 | -0.103 | 0.042 | - |

*Table 3: Correlation analysis of the survey (Panel A) and Amazon review (Panel B) datasets*

The correlation results for the survey variables (Panel A) show clear correlation between review reliance, credibility perceptions, and demographics. Review dependence is positively correlated with trust in verified reviews (r = .444), trust in detailed reviews (r = .365), and sensitivity to negative reviews (r = .466), indicating that consumers who rely more on reviews also trust credibility cues and are more affected by negative content. Credibility dimensions cluster together, as trust in detailed reviews is associated with trust in verified reviews (r = .364) and verified plus detailed reviews (r = .200). Sensitivity to negative reviews also links positively with credibility measures, reinforcing ties between negativity bias and trust. Age correlates negatively with review reliance (r = –.130), trust in verified and detailed reviews (r = –.212), and sensitivity to negative reviews (r = –.293), showing younger respondents report higher reliance and sensitivity. Purchase frequency is negatively associated with review dependence (r = –.191) but positively with age (r = .334), suggesting demographic influences on purchasing behaviour.

For Amazon reviews (Panel B), review helpfulness correlates positively with review length (r = .084), showing longer reviews receive more helpfulness votes. Verified purchase status is negatively related to review length (r = –0.413), suggesting unverified reviews are longer, while verified ones are shorter but impactful. A strong positive relationship exists between sentiment score and star ratings (r = .659), confirming alignment between text sentiment and rating. Verified purchase status is weakly but positively associated with review helpfulness (r = .067), though it correlates negatively with star ratings (r = –.103).

### 4.3 Regression Analysis

Regression analyses were conducted to test the proposed hypotheses across both survey and Amazon review data. Models were specified with relevant independent, dependent, moderator, and control variables to assess the effects of credibility cues, review detail, and sentiment. Interaction terms were included to evaluate moderation, while Variance Inflation Factors (VIF) were reported to confirm that multicollinearity was within acceptable limits.

#### 4.4.1 Regression Analysis of sentiment and star ratings for consistency on Amazon reviews

| Variable | B | Std. Error | t | Sig | VIF |
|---|---|---|---|---|---|
| constant | 4.2184 | 0.0283 | 149.1646 | 0 | |
| ***Review sentiment*** | 0.9274 | 0.0309 | 30.0097 | <0.001 | 1.015 |
| ***Review length*** | 0.0008 | 0.0002 | 4.083 | <0.001 | 1.217 |
| ***Verifies purchase status*** | -0.0595 | 0.0431 | -1.3801 | 0.1676 | 1.22 |
| | | | | | |
| **Avg VIF** | 1.151 | | | | |
| **Max VIF** | 1.22 | | | | |
| | | | | | |
| **$R^2$** | 0.441 | | | | |
| **F** | 300.468 | | | | |
| **p** | <0.001 | | | | |
| Dependent variable: Rating of reviews | | | | | |

*Table 4: Regression analysis on review sentiment and star rating*

The regression analysis tested whether review sentiment predicts star ratings, while controlling for review length and verified purchase status. As shown in Table 4, the model was statistically significant, with an $R^2$ of 0.441, indicating that approximately 44.1% of the variance in star ratings was explained by the predictors.

Review sentiment was a strong and significant predictor of star ratings ($B = 0.927$, $p < .001$), showing that more positive sentiment scores were associated with higher ratings. Review length also showed a small but significant effect ($B = 0.0008$, $p < .001$), suggesting longer reviews were more strongly linked with higher ratings. In contrast, verified purchase status was not a significant predictor ($B = -0.059$, $p = 0.168$), indicating no consistent difference in ratings between verified and unverified reviews. The variance inflation factors (VIFs) were low with Max VIF = 1.22, confirming no multicollinearity among the variables. Therefore, hypothesis H1 is valid.

### 4.4.2 Regression of Trust in verified reviews and Helpfulness

**Panel A – Survey**

| Variable | B | Std. Error | t | Sig | VIF |
|---|---|---|---|---|---|
| (constant) | 4.1815 | 0.0405 | 103.2294 | 0 | |
| ***Review reliance*** | 0.4166 | 0.0511 | 8.1463 | <0.001 | 1.122 |
| ***Purchase frequency*** | -0.0296 | 0.0447 | -0.6634 | 0.5076 | 1.164 |
| ***Review reliance × Purchase frequency*** | -0.0277 | 0.0461 | -0.6004 | 0.5487 | 1.119 |
| ***Age*** | -0.002 | 0.0061 | -0.3327 | 0.7396 | 1.14 |
| **Avg VIF** | 1.136 | | | | |
| **Max VIF** | 1.164 | | | | |
| **$R^2$** | 0.2 | | | | |
| **F** | 18.843 | | | | |
| **p** | <0.001 | | | | |

Dependent Variable: Trust in Verified reviews

*Table 5: Regression of Trust in Verified Reviews on Review Reliance*

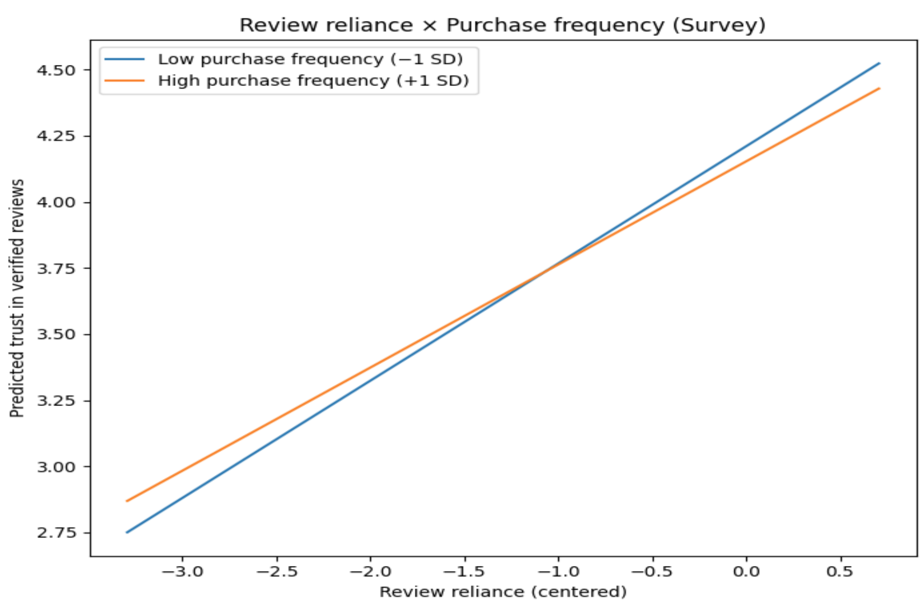

*Figure 1. Interaction plot of Review Reliance × Purchase Frequency on Trust in Verified Reviews*

The regression model tested the effect of review reliance on trust in verified reviews, with purchase frequency as a moderator and age as a control as shown in table 5. The model was statistically significant, with an $R^2$ of 0.200, indicating that 20% of the variance in trust in verified reviews was explained. Review reliance showed a strong positive effect (B = 0.4166, $p < .001$), while purchase frequency (B = -0.0296) and the interaction between review reliance and purchase frequency (B = -0.0277) were not significant. Age, included as a control variable, also showed no significant effect. VIF were all low (Max VIF = 1.164), indicating no multicollinearity.

Figure 1 shows the interaction between review reliance and purchase frequency on trust in verified reviews. The slopes for both high and low purchase frequency groups are parallel, indicating that while review reliance consistently increases trust in verified reviews, the moderating effect of purchase frequency is minimal. This suggests that consumers who rely more heavily on reviews show greater trust in verified reviews regardless of their purchase frequency.

**Panel B - Amazon**

| Variable | B | Std. Error | t | Sig | VIF |
|---|---|---|---|---|---|
| (constant) | 0.1011 | 0.0179 | 5.6493 | <0.001 | 1.201 |
| ***Verified purchase status*** | 0.4092 | 0.0377 | 10.8536 | <0.001 | 3.256 |
| ***Review rating*** | 0.0367 | 0.0186 | 1.9773 | 0.048 | 1.199 |
| ***Review length*** | 0.1979 | 0.0157 | 12.6139 | <0.001 | 3.285 |
| ***Verified × Rating*** | -0.0369 | 0.0244 | -1.5126 | 0.1304 | |
| **Avg VIF** | 2.235 | | | | |
| **Max VIF** | 3.285 | | | | |
| **$R^2$** | 0.112 | | | | |
| **F** | 42.126 | | | | |
| **p** | <0.001 | | | | |

Dependent Variable: Review helpfulness

*Table 6: Regression of Trust in Verified Reviews on Review Reliance*

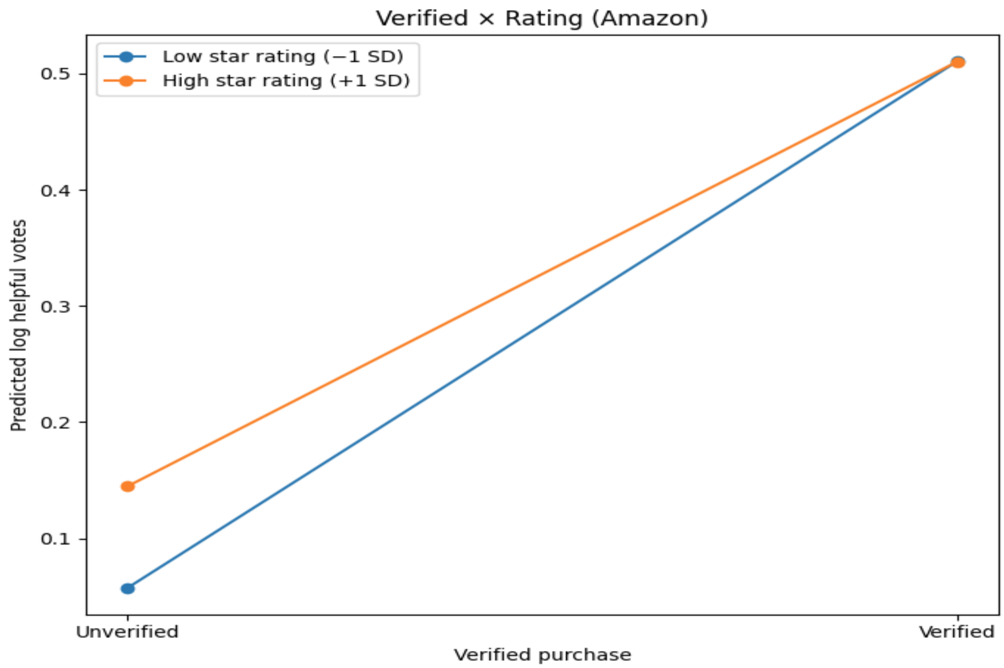


*Figure 2. Interaction plot of Verified Purchase × Star Rating on Review Helpfulness*

The regression model tested review helpfulness as the dependent variable, predicted by verified purchase status, star rating, and review length, with an interaction term between verified purchase and rating as shown in table 6. The model was statistically significant, $F = 42.126$, $p < .001$, with an $R^2$ of 0.112, meaning that 11.2% of the variance in helpful votes was explained. Verified purchase status was a strong positive predictor of helpfulness (B = 0.4092, $p < .001$). Review length also showed a significant positive effect (B = 0.1979, $p < .001$), while star rating was weaker but statistically significant (B = 0.0367, $p = .048$). The interaction between verified purchase and rating was not significant (B = -0.0369). with Max VIF = 3.285 there is no multicollinearity.

Figure 2 illustrates the interaction plot of verified purchase and star rating on predicted helpful votes. The plot shows that verified reviews consistently received more helpfulness votes than unverified ones across both low (−1 SD) and high (+1 SD) star ratings. While the slopes differ slightly, the non-significant interaction term suggests that the credibility benefit of verification does not vary substantially across different rating levels.

**4.4.3 Regression on trust in Detailed reviews on Review Reliance and Review Helpfulness.**

**Panel A – Survey**

| Variable | B | Std. Error | t | Sig | VIF |
|---|---|---|---|---|---|
| constant | 4.1481 | 0.047 | 88.2656 | 0 | |
| ***Review reliance*** | 0.3703 | 0.0593 | 6.2405 | <0.001 | 1.122 |
| ***Purchase frequency*** | -0.0233 | 0.0518 | -0.4499 | 0.6531 | 1.164 |
| ***Review reliance × Purchase frequency*** | 0.0312 | 0.0535 | 0.5827 | 0.5605 | 1.119 |
| ***Age*** | 0.0032 | 0.0071 | 0.4463 | 0.6557 | 1.14 |
| **Avg VIF** | 1.136 | | | | |
| **Max VIF** | 1.164 | | | | |
| **$R^2$** | 0.135 | | | | |
| **F** | 11.798 | | | | |
| **p** | <0.001 | | | | |

Dependent variable: Trust in detailed reviews

*Table 7: Regression analysis on Trust in Detailed Reviews on Review Reliance*

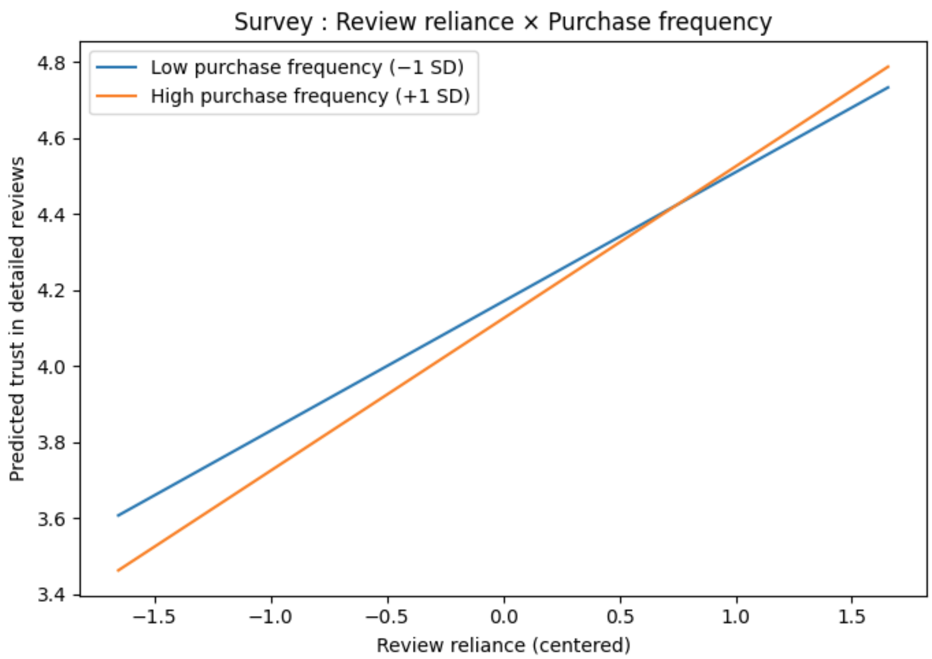

*Figure 3. Interaction plot of Review Reliance × Purchase Frequency on Trust in Detailed Reviews*

Table 7 presents the regression results for the survey data, where *Trust in Detailed Reviews* was regressed on *Review Reliance*, *Purchase Frequency* (moderator), and *Age* (control). The results indicate that review reliance had a strong and significant positive effect (B = 0.3703, $p < 0.001$), suggesting that greater reliance on reviews is associated with higher trust in detailed reviews. However, the interaction term between review reliance and purchase frequency was not significant, indicating that purchase frequency did not moderate this relationship. The overall model explained 13.5% of the variance in trust in detailed reviews.

As shown in figure 3, the interaction plot shows the relationship between review reliance and trust in detailed reviews under low and high purchase frequency. The plot shows parallel slopes, indicating that while review reliance consistently increases trust in detailed reviews, the moderating effect of purchase frequency is limited. Both low- and high-frequency purchasers follow similar trajectories, confirming the non-significant interaction effect seen in the regression model.

**Panel B - Amazon**

| Variable | B | Std. Error | t | Sig | VIF |
|---|---|---|---|---|---|
| constant | 0.1464 | 0.0171 | 8.5485 | 0 | |
| ***Verified Purchase Status*** | 0.3907 | 0.0348 | 11.2156 | <0.001 | 1.38 |
| ***Review Length*** | 0.1364 | 0.0221 | 6.1584 | <0.001 | 2.418 |
| ***Review rating*** | 0.014 | 0.0124 | 1.1333 | 0.2571 | 1.019 |
| ***Verified Purchase × Log Review Length*** | 0.0998 | 0.0303 | 3.2915 | 0.001 | 2.807 |
| | | | | | |
| **Avg VIF** | 1.906 | | | | |
| **Max VIF** | 2.807 | | | | |
| | | | | | |
| **$R^2$** | 0.117 | | | | |
| **F** | 42.821 | | | | |
| **p** | <0.001 | | | | |

Dependent variable: Credibility of detailed reviews

*Table 8: Regression analysis on Trust in Review Helpfulness on Verified × Review Length*

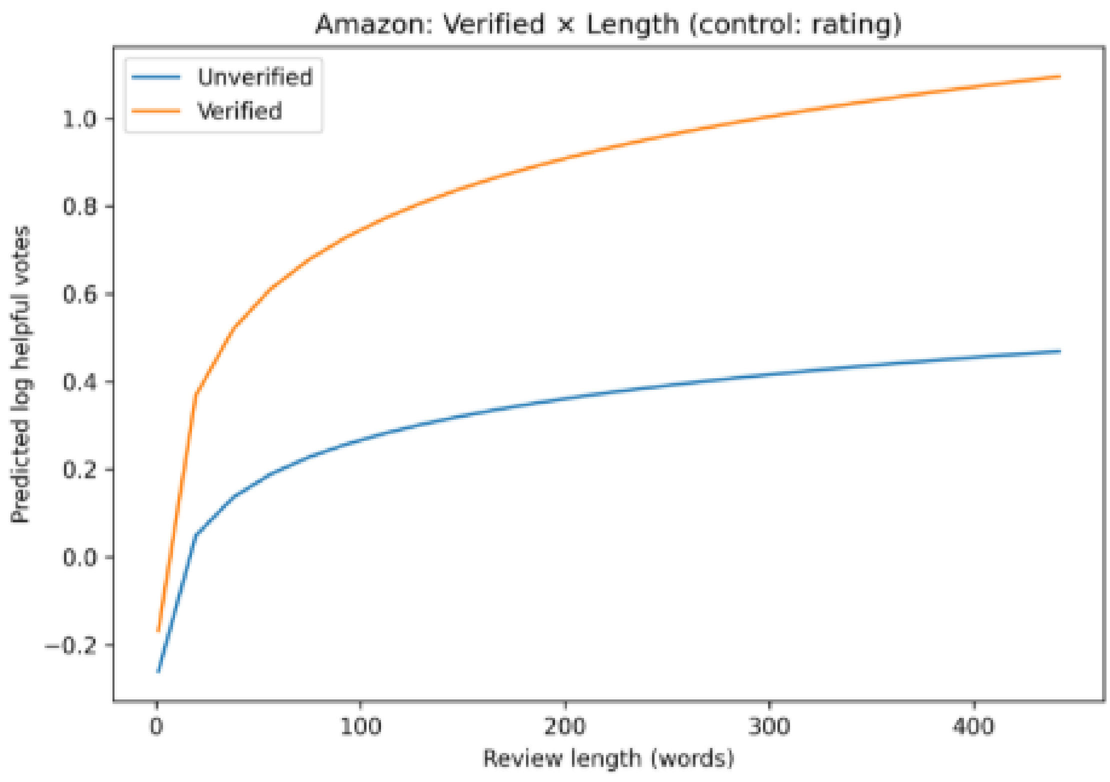


*Figure 4. Interaction plot of Verified Purchase × Review Length on Review Helpfulness*

Table 8 shows the regression results testing whether verified purchase status and review length (log-transformed) influence review helpfulness for Amazon data. Verified purchase status (B = 0.3907) and review length (B = 0.1364) both showed significant positive effects, indicating that verified reviews and longer reviews were more likely to be rated as helpful. Importantly, the interaction between verified purchase and review length was also significant (B = 0.0998), suggesting that the impact of review length on helpfulness was amplified when the review was verified. The model explained 11.7% of the variance in review helpfulness.

Figure 4 illustrates this interaction effect, showing that verified reviews consistently attracted higher predicted helpfulness votes compared to unverified reviews across different lengths. Moreover, the gap between verified and unverified reviews widened as review length increased, indicating that detailed verified reviews are particularly influential in driving perceptions of helpfulness

### 4.4.4 Regression of Negativity Bias in Review Helpfulness

**Panel A - Survey**

| Variable | B | Std. Error | t | Sig | VIF |
|---|---|---|---|---|---|
| const | 3.7919 | 0.0463 | 81.8256 | 0 | |
| ***Trust in credibility reviews*** | 0.2154 | 0.0636 | 3.386 | <0.001 | 1.229 |
| ***Review reliance*** | 0.4451 | 0.0599 | 7.4304 | <0.001 | 1.193 |
| ***Trust in credibility × Review reliance*** | 0.0289 | 0.0457 | 0.6318 | 0.528 | 1.232 |
| ***Age*** | -0.0283 | 0.0067 | -4.2187 | <0.001 | 1.07 |
| | | | | | |
| **Avg VIF** | 1.181 | | | | |
| **Max VIF** | 1.232 | | | | |
| | | | | | |
| **$R^2$** | 0.299 | | | | |
| **F** | 32.154 | | | | |
| **p** | <0.001 | | | | |

Dependent variable: Trust in negative review

*Table 9: Regression analysis on Sensitivity to Negative Reviews on Trust in Credibility Cues and Review Reliance*

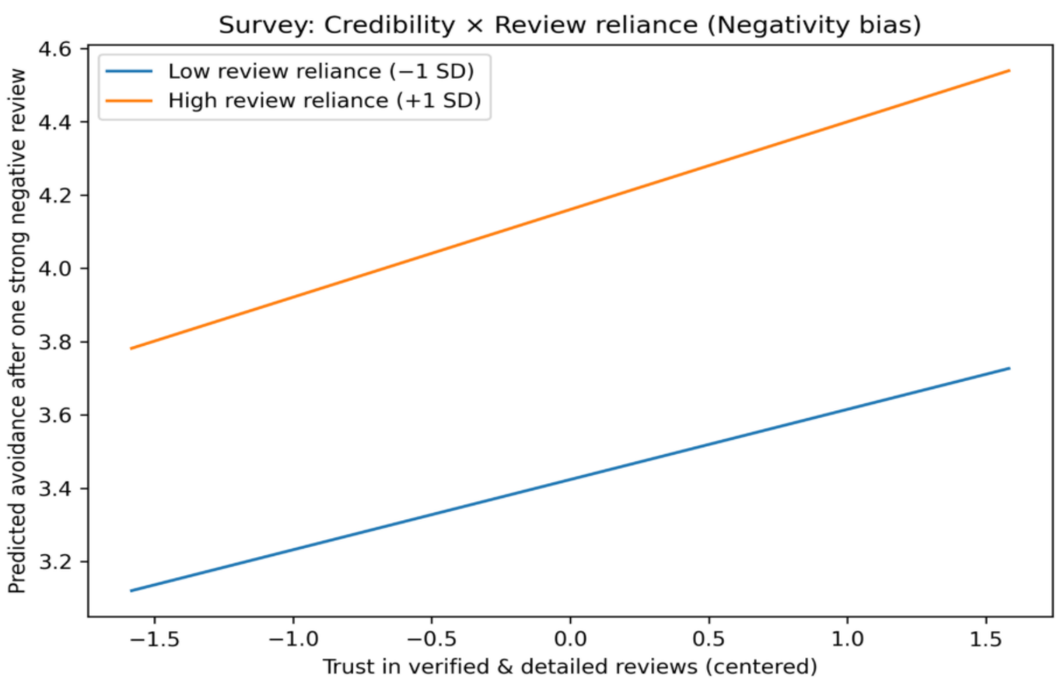

*Figure 5: Interaction plot of Trust in Verified and Detailed Reviews × Review Reliance on Negative Reviews*

Table 9 presents the regression results showing the relationship between trust in credibility cues, review reliance, and sensitivity to negative reviews, with age included as a control variable. The model shows a significant variance. Both trust in credibility cues ($B = 0.215$, $p < 0.001$) and review reliance ($B = 0.445$, $p < 0.001$) were significant predictors of sensitivity to negative reviews, suggesting that individuals who place greater trust in credible reviews and rely more heavily on them are more likely to report avoidance behavior after encountering a negative review. In contrast, the interaction between trust in credibility and review reliance was not significant. Age was also a significant negative predictor ($B = -0.028$), indicating that younger respondents were more sensitive to negative reviews than older ones. There is no multicollinearity with all VIF values below 1.3.

Figure 5 illustrates the interaction plot of credibility × review reliance, showing that sensitivity to negative reviews increases consistently with higher trust in credibility and stronger review reliance. Respondents with high reliance and high credibility trust demonstrated the greatest predicted avoidance after encountering negative reviews, while those with lower reliance reported less sensitivity. Although the interaction effect was non-significant in the regression, the plot highlights the consistent upward trend across reliance levels.

**Panel B - Amazon**

| Variable | B | Std. Error | t | p | VIF |
|---|---|---|---|---|---|
| constant | 0.2018 | 0.0172 | 11.7628 | 0 | |
| ***Negative sentiment*** | 0.0181 | 0.0221 | 0.8158 | 0.4146 | 3.152 |
| ***Review length*** | 0.001 | 0.0003 | 3.1575 | <0.001 | 1.48 |
| ***Verified purchase status*** | 0.3973 | 0.048 | 8.2848 | <0.001 | 1.489 |
| ***Review rating*** | 0.0122 | 0.0169 | 0.7223 | 0.4701 | 1.87 |
| ***Negativity × Review length*** | -0.0001 | 0.0003 | -0.4344 | 0.664 | 1.732 |
| ***Negativity × Verified purchase*** | -0.0633 | 0.0588 | -1.0764 | 0.2817 | 3.745 |
| ***Review length × Verified purchase*** | 0.0044 | 0.0009 | 5.0263 | <0.001 | 1.754 |
| ***Negativity × Review length × Verified purchase*** | -0.0015 | 0.0011 | -1.2981 | 0.1942 | 2.761 |

| | |
|---|---|
| **Avg VIF** | 2.248 |
| **Max VIF** | 3.745 |
| | |
| **R²** | 0.115 |
| **F** | 10.193 |
| **p** | <0.001 |

Dependent variable: Review Helpfulness

*Table 10: Regression analysis on Review Helpfulness on Negative Sentiment × Verification × Review Length*


*Figure 6: Interaction plot of Negativity × Review Detail × Verification on Review Helpfulness*

Table 10 presents the regression analysis examining the combined effects of negativity, review length, and verification status on review helpfulness, with rating included as a control variable. The results indicate that review length ($B = 0.0010$, $p < 0.01$) and verified purchase status ($B = 0.3973$, $p < 0.001$) are significant positive predictors of helpful votes. However, negativity alone was not significant ($p = 0.4146$). Among the interaction terms, review length × verified purchase ($B = 0.0044$, $p < 0.001$) were significant, suggesting that the credibility effect of verification is amplified when reviews are longer. The full three-way interaction between negativity × length ×

verification was not statistically significant. The overall model fit was modest ($R^2 = 0.115$, $F = 10.193$, $p < 0.001$).

Figure 6 shows these results through interaction plots. For unverified reviews, negativity and detail level show minimal influence on helpfulness. However, in verified reviews, the plots reveal that review detail moderates the effect of negativity, with highly detailed negative reviews attracting more helpfulness than less detailed ones. This shows that verification and review depth together influence how helpful a review is seen, leading to place more trust in negative reviews.

# CHAPTER V DISCUSSIONS

## 5.1 Overview

This chapter discusses the findings of the study in relation to the research objectives, theoretical framework, and prior literature. The purpose of the dissertation was to understand the combined impact of sentiment, credibility cues, and negativity bias in online reviews on consumer trust and purchasing perceptions in the skincare sector. The results are explained through four hypotheses, compared with earlier studies on online shopping behaviour, and considered for both theory and real-world use. The chapter also discusses the study's limitations and possible other explanation.

## 5.2 Interpretation of Findings

The findings for H1 confirmed a strong positive relationship between review sentiment and star rating in the Amazon dataset. Reviews with more positive textual tone consistently mapped onto higher star ratings, with sentiment as the strongest predictor in the regression analysis. This aligns with prior studies demonstrating that textual valence is generally congruent with rating behaviour (Archak et al., 2011; Floyd et al., 2014). The association suggests that consumers do not randomly give ratings, they use their words in detail to show their satisfaction or dissatisfaction with the product. Interestingly, review length also contributed positively, indicating that longer reviews are slightly more associated with higher ratings. This could reflect a positivity bias in voluntary contributions, where satisfied consumers invest more effort in writing elaborate reviews (Mudambi & Schuff, 2010). Ratings are mostly shaped by how satisfied consumers feel and the tone of their review, while other credibility cues matter more when judging how helpful a review is, not when giving star ratings.

H2 explored the role of verified purchase status as a credibility cue and its moderating interaction with consumer reliance on reviews (survey) and star ratings (Amazon). In the survey model, review reliance was a significant predictor of trust in verified reviews, but the interaction with purchase frequency was non-significant. This suggests that consumers who generally depend on reviews shows higher trust in verified reviews, regardless of their frequency of purchases. While purchase frequency was theorised to strengthen this effect (i.e., frequent buyers would be more sceptical and selective), the results suggest that the "verified" label is seen as a universally strong signal, unaffected by shopping habits. In the Amazon model**,** verified purchase status significantly predicted helpfulness, aligning with prior work showing that badges improve credibility (Luca & Zervas, 2016; Chevalier & Mayzlin, 2006). Review length also strongly predicted helpfulness, supporting earlier evidence that longer reviews provide more useful information (Mudambi & Schuff, 2010). However, the expected interaction between verified status and star ratings was not significant. This contrasts with the hypothesis that consumers would value verified positive reviews more than unverified ones. Instead, the results suggest an additive heuristic, where verification and ratings influence helpfulness independently rather than jointly. Therefore, H2 is partially supported. Verification consistently boosts credibility, but its interaction with other factors was weaker than expected. This shows consumers may simplify their judgments by adding signals together instead of considering how they interact and treating every variable as independent edibility signal (Metzger et al., 2010; Filieri, 2015; Sundar, 2008).

H3 addressed whether detailed review increase trust and perceived helpfulness, the survey results, reliance on reviews significantly predicted higher trust in detailed reviews. However, the interaction with purchase frequency was not significant, like the pattern observed in H2. This suggests that consumers who rely on reviews tend to place greater trust in detailed content, regardless of how often they purchase. Detail therefore appears to function as a strong credible signal, consistent with the central route of the Elaboration Likelihood Model (Petty & Cacioppo, 1986), where elaboration in the message boosts credibility. In the Amazon model, review detail (measured by review length) was a significant predictor of credibility. The interaction between verified status and review length was also significant indicating that long verified reviews are seen most credible and useful over short, verified reviews. The results suggest that detail alone is not always sufficient. It is strengthened by verification. Overall, H3 was strongly supported, particularly in the Amazon model, which revealed clear interaction effects. Together, these results reinforce the uniqueness between H2 and H3: whereas H2 examined source credibility cues (verified purchase as a marker of authenticity), H3 focused on message credibility cues (review detail/length as a marker of informativeness). This separation confirms that credibility operates through different pathways; the credibility of the source versus the credibility of the message itself.

H4 examined whether consumers place more weight on negative reviews and how this effect is shaped by credibility. In the survey model, both trust in credibility cues and reliance on reviews had greater sensitivity to negative reviews. his supports the negativity bias hypothesis (Baumeister et al., 2001), showing that consumers who already trust and depend on reviews are more likely to avoid products after seeing negative feedback. However, the interaction between credibility and reliance was not significant, suggesting that these effects operate independently. Age also negatively predicted sensitivity, indicating that younger consumers are more influenced by negative reviews. In the Amazon model, negativity on its own was not a strong predictor. Instead, its influence depended on credibility and detail. Verified purchase status significantly increased helpfulness and review length also contributed positively. Importantly, verified, detailed negative reviews were judged especially impactful. his pattern supports the idea that negativity bias is conditional i.e. negative information becomes more persuasive when it comes from credible and effortful sources. Overall, H4 provides evidence that negativity bias exists but is not universal. Instead, it becomes stronger when paired with credibility signals, showing that negativity bias depends on how trustworthy and high-quality the review appears.

**5.3 Theoretical Contributions**

This study adds to the research on online reviews, credibility cues, and consumer decision-making. It combines survey-based attitudes with real behaviour data from Amazon. It improves our understanding in three main ways: it clarifies the role of credibility cues within the Elaboration Likelihood Model (ELM), it extends theories of source credibility and negativity bias, and it shows how different review signals such as sentiment, verification, length, and valence interact to shape up trust and credibility.

This study extends the Elaboration Likelihood Model (Petty & Cacioppo, 1986) by showing review attributes can act as central or peripheral cues depending on the context. Sentiment valence acts as a central cue as readers view reviews more deeply when the textual tone aligns with star rating. However, verification and review length act as additional cues showing that credibility is acquired without requiring extensive thought. Additionally, the results shows that cues can interact, such as detailed verified reviews supports both central and peripheral processing, showing that cue processing is context dependent rather than strictly hierarchical.

This study also advances source credibility theory (Hovland & Weiss, 1951) by showing how platform-generated cues (verification badges) interact with user-generated signals (sentiment and detail). Prior work studies these effects separately, but our findings show the effect of both additive and multiplicative effects. Verification status consistently increased trust and helpfulness, in line with past studies (Luca & Zervas, 2016). However, the moderation by star rating was not significant, suggesting that credibility often with independently supporting theories of cue accumulation (Metzger & Flanagin, 2013) that consumers sometimes apply additive heuristics than complex cue integration.

The study refines the theories of negativity bias (Baumeister et al., 2001). Survey results showed that consumers who rely heavily on reviews were especially sensitive to negative content, consistent with prior findings that negative cues outweigh positive ones (Ludwig et al., 2013). However, in amazon data the effect of negativity bias appeared only when reviews were both verified and detailed. This suggests that negativity bias is automatic but shaped by credibility signals connecting the negativity bias research with review credibility theory.

Finally, this study also makes a methodological contribution by combining survey responses with real behaviour from Amazon. This mixed approach provides stronger real-world validity and reliability, since it connects what people say (perception) with what they do (behaviour). Compared to studies using only one method, it gives a complete picture of how reviews influence consumer decisions.

### 5.4 Practical Contributions

The findings from the study provides multiple practical implication for platforms, brands and consumer engagement. Firstly, platforms should make their sentiment and ratings more visible by using algorithms to summarise overall sentiment, giving shoppers quick “emotional overviews” of products (Mudambi & Schuff, 2010), since star ratings often reflect the tone of the reviews showing both can help consumers to process information easily. Additionally, the verification badge remains essential. Verified status increases the perceived trust and credibility of the consumers by highlighting the importance of transparency. Review platforms should continue these badges. However, because verification and ratings do not always interact strongly. For high-risk purchases, verified badges should be more prominent, while for low-risk items, sentiment and ratings may be enough. Firms should also encourage for collecting detailed reviews, especially when it’s a verified purchase. Long verified reviews were seen as far more helpful than short or unverified ones. Firms could encourage this by rewarding verified customers for writing detailed reviews through loyalty points, discounts, or recognition. Platforms could also add prompts or filters to guide reviewers toward describing usage, benefits, and drawbacks, improving review quality.

Skincare brands need to manage their negative reviews strategically, when negative bias was present it was significant for verified detailed reviews rather than hiding these reviews, brands should respond proactively (Coombs, 2007). Public replies that acknowledge issues and offer solutions can reduce damage and show accountability. Platforms could add context to negative reviews, such as showing if the reviewer is a frequent buyer or highlighting reviews that include both pros and cons. Some changes to the policies can be made from this study, since unverified long reviews were less influential, platforms could tighten their verification policies to reduce fake, false or manipulative content this will protect the rust and reputation in the review system.

Finally, for marketers and managers, the findings provide the importance of integrating review analytics into reputation management. Tracking not only star ratings but also sentiment, verification, and review length can provide early warnings of shifts in consumer trust. Advanced tools combining natural language processing with review metadata could help identify high-risk negative reviews (Verifies, long negative reviews) and allow firms to respond quickly before reputational damage spreads.

# CHAPTER VI CONCLUSION

This chapter will summarise the key findings of this study. In addition, with its limitations and scope for future research will be discussed.

## 6.1 Key Findings of Study

This objective of this research is to understand the effect of sentiment, the attributes that contributes to credibility and the presence of negativity bias in online reviews on consumer perception and purchase intent, with a focus on skincare product reviews. To address these two datasets we collected, the primary dataset consisted of consumer survey with 307 responses collected from various anonymous respondents, which captures self-reported perception and attitude towards credibility cues such as trust in verified reviews, trust in detailed reviews, reliance on reviews before purchase, and sensitivity to negative reviews, alongside demographic measures like age and purchase frequency for the analysis of psychological dimensions of review credibility and consumer reliance on online information. The secondary dataset was from archived Amazon's skincare product reviews, a total of 2012 filtered reviews were obtained with variables including star ratings, sentiment scores (computed using a pre-trained NLP model), helpfulness votes, verified purchase status, and review length allowing the analysis of behavioural outcomes, specifically whether credible and emotional reviews attract greater trust from other consumers.

The study finds that credibility cues such as verified and detailed review that has higher ratings strongly influences customer trust in survey and more helpfulness in Amazon reviews, additionally the relationship between the sentiment and the star rating was strong and consistent among the reviews. The study also finds negative review has greater influence when it is accompanied by credibility markers such as verification and detail, providing clear evidence of negativity bias. The findings show that both survey perceptions and Amazon behaviours lead to same result where consumers consistently respond to credibility cues and sentiment cues in reviews.

This Study contributes by understand the consumer perceptions by self-reported questionnaire with real world review outcomes, offering multi-dimensional understanding of how online reviews shape consumer decision making. The study's findings extend the existing theory, especially on Elaboration Likelihood model (ELM) by showing how both central cues such as detail and sentiment and peripheral cue such as verification guides the perception and the understanding of the review by the consumers. The study also contributes with practical insights for e-commerce platforms and brands on the importance of credibility indicators and the need for structures that support detailed authentical consumer feedback.

### 6.2 Limitations and Recommendations for Future Research

While the study has meaningful contributions to both theory and practice, it has some limitations that must be acknowledged. Firstly, the total number of respondents that was considered for this research is not large enough, only 307 survey responses were collected through the survey questionnaire. While it provides reasonable foundation for regression and correlation analysis, the modest sample size reduces the statistical power of the model. The secondary dataset that was used was from archived amazon dataset that had only 7000 reviews in which 2012 reviews were filtered for the analysis based on keywords. This may affect the insight on consumer behaviour to full extent. Additionally, the study did not differentiate the results based on platforms (Amazon vs other platforms) or by brand or it's sub products, which reduces the degree of contextual relevance. For example, reviews of luxury skincare brands may have more weight on detail and verification, while mass- market brands rely on just verification cues. By not considering these multiple attributes, this study explores broad patterns of credibility but cannot fully explain brand or platform specific dynamics in skincare reviews.

Adding on the limitations, this study has several directions for future research that further improves the understanding of credibility and customer perception. Firstly, improving the statistical power of the model by increasing the sample size for better accuracy and insight (Cohen, 1992). Additionally, using a larger and more samples across different industries or brands would make the findings more generalised and test whether the findings hold beyond skincare. Study on longitudinal data over long periods will help us understand how consumer trust, sentiment and perception changes over time and evolve with context (Baek, Ahn, & Choi, 2012). The study can extend the work on analysing the brand and product differences for example, whether luxury and mass-market brands have different responses to credibility cues or does the sentiment differs from platform to platform providing insight and targeted policies changes for platforms. The study uses DistilBERT for sentiment analysis which improves accuracy over lexicon-based tools like

VADER (Hutto & Gilbert, 2014). These models can miss cultural nuances, irony or sarcasm, creating a machine learning model with manual coding will greatly improve the validity of the findings (Thelwall, 2018). Finally, including reviewer expertise, social identity and other multimodal data such as images, videos etc. can provide more structure on how consumers perceive the review data.

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# APPENDIX I - ETHICAL APPROVAL DOCUMENT

University of Bristol

## University of Bristol Research Ethics Application

### Investigator information

Application Submitter Details

Title

Mr

First Name

Gokul

Surname

Kumar

Faculty

Arts, Law and Social Sciences

Department

School of Management - Business School

School

Telephone

Email

ha24467@bristol.ac.uk

Preferred Name or Also Known As

26 August 2025

Reference #: 2025-26957-29292

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Faculty

Social Sciences and Law

School / Department / Centre

University of Bristol Business School

Are you a student submitting this ethics application as part of your degree qualification?

Yes

Please declare your level of study

Taught Masters

Supervisor Contact Details

Title

Dr

First Name

Tian

Surname

Han

Department

Management

Faculty

Faculty of Social Sciences and Law

Email

tian.han@bristol.ac.uk

Business School Academic Ethics Checker

Title

Dr

First Name

Xin

Surname

Fei

Department

School of Management - Business School

Faculty

Arts, Law and Social Sciences

Email

qc20250@bristol.ac.uk

Are you an academic member of staff submitting an ethics application on behalf of a student(s) as part of their degree qualification?

No

Second Supervisor Details. If University of Bristol, please provide their full name and title.

If external to the University of Bristol, please provide their name, organisation details, email address and telephone number.

N/A

Please provide details of any other researchers/collaborators involved in the study.

N/A

Are you submitting this ethics application on behalf of another researcher?

No

### Ethics Committee Review

Has or will your research be submitted to another research ethics committee for research involving human participants, their tissue and or data?

- ☐ Yes
- ☑ No

**Important Information - Please note:**

It is extremely important that you select the **correct Research Ethics Committee (REC)** to review your research ethics application.

The REC selected, will determine the questions you are asked to complete on this online form and the research ethics committee that will review your research ethics application.

**Please note**, if you select the incorrect ethics committee, this may delay the review of your ethics application as your ethics application will need to be returned to you so that you can select the correct REC and complete the relevant questions on the online form.

If you are unsure of the correct research ethics committee to select please contact research-ethics@bristol.ac.uk

Please select the Research Ethics Committee (REC) to review your research ethics application:

Business School Research Ethics Committee

To proceed to the next page select 'Next' in the Actions tiles.

To save your application for completion and submission at a later date please select 'Save' in the Actions tiles.

### Lead questions

Does your research involve any of the following? Tick all that apply

- ☐ Patients, clients or carers of an NHS Trust or Social Care organisation
- ☐ Solely staff or premises of an NHS Trust or Social Care organisation?
- ☐ Staff, residents or premises of one or more care homes
- ☐ Participants who are lacking or have diminished mental capacity
- ☐ Staff, inmates or premises of one or more prisons
- ☐ Staff, participants or premises of one or more local authority departments
- ☑ None of the above

To proceed to the next page select 'Next' in the Actions tiles.

To save your application for completion and submission at a later date please select 'Save' in the Actions tiles.

## Brief study outline

Study Title

How Online Review Credibility and Negative Bias Influence Consumer Trust in Skincare Brands

Brief Project Outline (up to approximately 300 words in plain English)

Many people rely on online reviews when choosing skincare products, as these products affect personal health and appearance. Some reviews seem more trustworthy than others, especially when they are detailed, balanced, and written clearly. On the other hand, very negative reviews, particularly those written in an emotional or one-sided manner, can often have a more substantial impact, even if the majority of reviews are positive. This study aims to investigate what factors contribute to the credibility of a review and how strongly negative reviews can impact people's perceptions of a skincare brand.

The project uses two methods to answer this question. First, I will analyse thousands of skincare product reviews taken from a large, publicly available Amazon dataset (created by McAuley Lab). This dataset includes anonymised customer reviews for a wide range of skincare products across different brands. Using tools like Python and natural language processing (NLP), I will look for signs of trustworthiness, such as verified purchase status, review length, spelling and grammar, and helpfulness votes, as well as signs of negative bias, including emotional tone, strong complaints, or blame.

Second, I will run a short online survey using Microsoft Forms. The survey will be shared with adults who buy or research skincare online. Participants will read sample reviews and answer questions about how much they trust them and whether they would still consider buying the product. The survey will take around 8–12 minutes to complete.

Together, these two approaches will help us understand how review features affect consumer trust. The findings will offer valuable insights for brands, online platforms, and future research into how digital opinions influence real-world decisions.

Estimated dates of research data collection

Please note that you must not start data collection for your research project until you have received a formal favourable ethics opinion from the appropriate Research Ethics Committee (REC). Please factor in the research ethics review timescales when selecting the estimated dates for research data collection.

Anticipated Start Date

01/07/2025

Anticipated End Date

31/07/2025

To proceed to the next page select 'Next' in the Actions tiles.

To save your application for completion and submission at a later date please select 'Save' in the Actions tiles.

## Data collection method

**BUSINESS SCHOOL RESEARCH ETHICS**
**APPLICATION FORM: TAUGHT POSTGRADUATE STUDENTS**

This form must be completed for each piece of research carried out by all taught post-graduate students in the Business School.

Students should discuss their proposed research with their supervisors who will then approve and sign this form before forwarding to the relevant dissertation convenor (or in some cases unit convenor or programme director) who will approve the form on behalf of the Business School REC when they are happy with the contents.

**Failure to get approval prior to conducting any fieldwork** may result in the University taking action for research misconduct – the outcome of such action may be that you are unable to submit your fieldwork findings for assessment and **your degree may not be awarded.**

Once your study is approved, you must follow the plan described in this form. You should remember that ethics is an on-going process, ie your ethical thinking is not 'done' when your form is signed. It is about how you act as a researcher. You should remain reflexive throughout the research process and think about how the research is impacting on your participants and yourself. You should refer to this completed form throughout your research process to make sure you are remaining within your ethical approval. If you wish to change your research plan, then you must discuss this with your supervisor. If the change is very small your supervisor can approve the change. However, if the change is more significant, you will need to ask for an amendment to your ethical approval. Your supervisor and dissertation convenor must approve this change in writing. If you do not get approval for changes, then you won't have ethical approval for the change, and it may result in the University taking action for research misconduct.

**Terminology used in this form:**

1. **Primary research** includes any research that collects new data such as interviews, focus groups, observations, online surveys, new data collected via a social media post etc.
2. **Secondary analysis/literature review** relates to the re-analysis of data that already exists such as analysis of publicly available documents or tv programmes, analysis of existing social media posts, reviews systematic or otherwise, or statistical analysis of analysis of publicly available datasets etc.

Please select the method of data collection relevant to your research. Tick all that apply

- [x] Primary research data collection
- [ ] Secondary data analysis

Please provide your student number

2608710

To proceed to the next page select 'Next' in the Actions tiles.

To save your application for completion and submission at a later date please select 'Save' in the Actions tiles.

## Study design and background

Research involving humans by all academic and related Staff and Students in the University of Bristol Business School is subject to the standards set out in the University of Bristol Ethics of Research Policy and Procedure which can be found at: http://www.bristol.ac.uk/red/research-governance/practice-training/researchethicspolicy.pdf

It is a requirement prior to the commencement of all funded and non-funded research that this form be completed and submitted to the School's Research Ethics Committee (REC). The REC will be responsible for issuing certification that the research meets acceptable ethical standards and will, if necessary, require changes to the research methodology or reporting strategy. It is a requirement that prior to the commencement of all funded and non-funded research that this form be completed and submitted to the School's Research Ethics Committee (REC). The REC will be responsible for issuing certification that the research meets acceptable ethical standards and will, if necessary, require changes to the research methodology or reporting strategy.

1 Background and aims of the research.

Online product reviews are a key influence on consumer decisions in the skincare industry, where product trust, effectiveness, and personal experiences are highly subjective. With the continued growth of e-commerce and digital word-of-mouth, consumers increasingly rely on the reviews and experiences of others to shape their perceptions of skincare brands and their products. This makes it important to examine the features of online reviews that foster trust and influence brand perception.

While prior research has explored the role of sentiment in affecting purchase intentions and brand attitudes (Babic Rosario et al., 2016), and the impact of credibility on consumer decision-making (Cheung et al., 2009), fewer studies have examined the combined influence of credibility cues (such as review length, grammar quality, verified purchase status, and helpfulness votes) and negative bias (the disproportionate weight given to strongly negative reviews), particularly within the skincare domain. Given the emotional and personal nature of skincare, elements like tone, authenticity, and experiential detail can significantly shape brand perceptions.

This study aims to investigate how credibility indicators and negatively biased review content impact consumer perception of skincare brands. A mixed-method approach will be used:

An online survey (via Microsoft Forms) to assess how consumers evaluate sample reviews and which features shape their perceptions of trust and brand value.

Textual analysis of a large-scale Amazon review dataset (collected by McAuley Lab), focusing on skincare reviews across a set of major skincare brands, selected based on data availability.

The findings will contribute to academic literature on consumer psychology and digital trust, while providing practical insights for skincare brands and platforms in managing their online reputation and mitigating the impact of negative user-generated content.

2 Outline the design of the study and list the procedures/activities to which the participants will be subjected:

This study adopts a mixed-methods design, combining primary data collection via an online survey with secondary analysis of a large-scale, pre-collected Amazon review dataset. The research aims to examine how credibility cues and negatively biased review characteristics influence consumer perception of skincare brands.

Primary data will be gathered through a structured online survey distributed via Microsoft Forms. The target participants are adults who have previously purchased or considered purchasing skincare products online. The survey will include Likert-scale items assessing perceived trustworthiness and influence of various review features, such as verified purchase status, review length, grammar, emotional tone, and helpfulness votes. These items will be followed by multiple-choice questions to explore which review characteristics most shape participant judgments. The survey is expected to take approximately 8–12 minutes to complete. No personally identifiable information will be collected, participation is fully anonymous, and respondents may withdraw at any point before submitting their responses.

Secondary data for this study will be drawn from the publicly available Amazon Product Reviews dataset compiled by McAuley Lab at the University of California, San Diego. This dataset spans reviews from May 1996 to September 2023 and includes anonymised user-generated content such as review text, star ratings, verified purchase indicators, and helpfulness votes. It is legally accessible for academic research and does not contain any personal or sensitive information.

This design enables an ethical, scalable, and data-rich approach to understanding how consumers interpret online skincare reviews. The combination of survey responses and review text analysis enables triangulation between consumer perceptions and real-world review patterns, providing both academic insights and practical guidance for digital marketing and brand management in the skincare industry.

3 Is this research project funded?

No

4 Do any of the investigators have any actual or potential conflict of interest in this study?

- [ ] Yes
- [x] No

## Research Location and Approvals

5 If participants are taking part without their knowledge or consent at the time, what steps will be put in place to ensure that participants are informed that the research has taken place? Will participants be debriefed about the covert nature of the research and given the opportunity to withdraw their data?

N/A

6 Where will the research data collection take place? Please be as specific as possible and if any access approvals are required.

The research data collection will take place entirely online. Primary data will be collected via a structured online survey created and distributed through Microsoft Forms. The survey link will be shared digitally through personal networks, social media platforms, and relevant online communities focused on skincare, to reach adult consumers with experience in purchasing skincare products online.

Secondary data will be obtained from the publicly available Amazon Product Reviews dataset compiled by McAuley Lab at the University of California, San Diego. This dataset was collected in 2023 and contains anonymised product reviews from Amazon, covering the period from May 1996 to September 2023. The dataset is openly accessible to the academic research community and comprises only non-identifiable user-generated content, including review text, star ratings, helpfulness votes, and verified purchase indicators.

No physical location or institutional access is required for this research. All data collection will be conducted remotely using publicly available tools and resources. No approvals are required to access the pre-collected dataset, and no personal or sensitive information will be collected during either phase of the study.

7 Have or will appropriate site approvals / letters of access be obtained before the research is undertaken?

- ☐ Yes
- ☐ Yes (In progress)
- ☑ No (None required)

If available for review, please upload any site approvals / letter of access from a site where you wish to undertake your research for review.

(E.g. confirmation from a school Principal that you can undertake your research data collection at their school)

## Participant recruitment

8 Who will be recruited to participate in the research? (Please be as specific as possible)

The participants for this research will be adults aged 18 and above who have experience purchasing or researching skincare products online. The survey will target general consumers, including both men and women, across a wide age range, but will primarily focus on young adults and middle-aged individuals (ages 18–45), as they represent the most active online shoppers of skincare products.

9 How many participants will be recruited? Provide justification for the sample size.

A minimum of 300 participants will be recruited for this study to ensure sufficient statistical power for quantitative analysis, including correlation and regression techniques. This sample size is justified based on existing literature recommendations for behavioural survey research where predictive modelling (e.g. multiple regression) is applied (Comrey & Lee, 1992; Tabachnick & Fidell, 2013).

Given that the survey will be distributed digitally via Microsoft Forms and shared across multiple social media channels and networks, the expected maximum sample size is approximately 500 participants. This range (300–500) provides flexibility while still maintaining a manageable dataset for analysis, interpretation, and ethical data handling.

10 How will the participants be identified and recruited to take part in your research?

Participants will be identified and recruited through non-probability convenience sampling. The survey link (hosted on Microsoft Forms) will be shared online via personal and professional networks, including WhatsApp, Instagram, and Facebook, as well as relevant skincare and beauty-related online communities and university peer groups. A digital recruitment poster will accompany the survey link in these channels and may also be circulated via email to eligible participants.

11 Are there any potential participants who will be excluded? If so, what are the exclusion criteria?

Yes. Participants under the age of 18 will be excluded. Additionally, individuals without experience in purchasing or reading online skincare product reviews will be asked to exit the survey early, as their input would not be relevant to the research aims.

12 If participants are receiving financial inducement (other than reasonable expenses and compensation for time), please provide details of the amount offered and justification for the amount.

N/A

13 Have you provided copies of all recruitment material with your ethics application for review?

- [x] Yes
- [ ] No
- [ ] N/A

Please provide any recruitment material used to recruit potential participants to take part in your research for review.

## Informed consent

14 If you are recruiting participants who are particularly vulnerable or unable to give informed consent, please detail how and from who, informed consent will be obtained.

N/A

Please provide copies of any participant facing study documentation used to inform participants about the nature of the research for review. Such as:

- Participant Information Sheet (PIS)
- Parental / Guardian Information Sheet (P/GIS)
- Child Assent Information Sheet

15 Clearly outline how informed consent will be obtained from all participants and / or their parents / guardians prior to individuals entering the research study?

Informed consent will be obtained through the first section of the Microsoft Forms survey. Participants will first be shown a detailed Participant Information and Consent Form outlining the study's purpose, voluntary nature, anonymity, and their right to withdraw at any point before submission.

Participants must actively tick a box confirming they have read and understood the information and agree to take part in the study before accessing any survey questions.

As the study targets adults aged 18 and above only, no parental or guardian consent is required.

16 How much time will participants be given to decide whether to give consent to participate after having been fully informed?

Participants will be given as much time as they need to review the Participant Information and Consent Form before proceeding. Since the survey is self-administered and accessible online, there is no time pressure, and participants can choose to start, pause, or exit the form at any point before submission.

17 Has an independent named contact been provided if a participant wished to make a complaint or raise any concerns regarding any ethical issues with the research?

Participants should be informed that they can contact the Research Governance Team (research-governance@bristol.ac.uk) as an independent contact if they wish to make a complaint or raise any concerns with this research.

- [x] Yes
- [ ] No

18 If your research involves the discussion or collection of information on sensitive topics that fall under special category status, do you confirm that you will obtain explicit consent for this and that you have provided your consent form / consenting statement for review?

- [ ] Yes
- [ ] No
- [x] N/A

19 Will participants be kept informed of new information that becomes available during the study which may influence their continued participation?

N/A

20 Will participants be made aware they can withdraw their person or data from the research study at any time without having to give a reason for doing so?

- [x] Yes
- [ ] No

20.1 If yes, outline the withdrawal process participants will need to follow to withdraw their person and / or data from the study.

Participants can withdraw from the study at any point before submitting the survey simply by closing the browser or exiting the Microsoft Forms page. However, once the survey is submitted, data cannot be withdrawn because responses are collected anonymously and cannot be linked back to individual participants. This will be clearly stated in the Participant Information and Consent Form.

20.2 Will the research data be fully anonymised? If so, will participants be made aware that they can request to withdraw their data at any time, but that due to data anonymisation, it may not be possible to comply with this request?

Yes, the research data will be fully anonymised. Participants will be clearly informed in the Participant Information and Consent Form that:

1. No personally identifiable information will be collected.

2. Responses are fully anonymous, meaning data cannot be linked back to individual participants.

Therefore, while participants may withdraw from the study at any time before submitting the survey, it will not be possible to withdraw their data after submission due to the anonymised nature of the data.

Please provide copies of any documentation such as Consent Forms used to obtain consent from participants before taking part in your research.

Documents

| Type | Document Name | File Name | Version Date | Version | Size |
|---|---|---|---|---|---|
| Consent Form | Participant Information and Consent Form (1) | Participant Information and Consent Form (1).docx | | | 61.8 KB |

## Participant and Researcher Safety

25 Are there any benefits to participants in taking part?

- Be clear and realistic – if there are no direct benefits for the participant, then please state this.

There are no direct personal benefits for participants in taking part in this study.

26 Describe potential risks to **research participants** (physical, psychological, legal, social) arising from the research and outline the processes implemented to mitigate against the identified risks:

This study involves no risk to participants. The research consists solely of an anonymous online survey about participants' opinions and experiences with online skincare reviews. No sensitive or personal questions are asked.

27 Describe potential risks to the **researcher** (physical, psychological, legal, social) arising from the research and outline the processes implemented to mitigate against the identified risks:

There are no significant physical, psychological, legal, or social risks associated with this study for the researcher.

## Data management and information security

30. If your research involves photographs, video, audio recording or similar of research participants, do you confirm that you will act in accordance with the relevant laws and University Information Security Policies?

- [ ] Yes
- [x] Not applicable

32 What arrangements have been put in place to ensure confidentiality and security of data gathered in the study? Will the data be stored in hard copy or electronically, and where will it be held?

All data collected in this study will be stored electronically and securely. Survey responses will be collected using Microsoft Forms, which automatically stores data in the researcher's University of Bristol OneDrive account. This storage is password-protected and accessible only to the researcher.

The secondary review data will be sourced from the publicly available Amazon Reviews dataset compiled by McAuley Lab. This dataset is already anonymised and does not contain any personal or identifiable information. It will be downloaded and stored as structured text files (e.g. JSON or CSV format) on the same secure OneDrive account.

No physical or hard copy data will be collected or stored. All information will be handled in accordance with University of Bristol data management and confidentiality policies.

## Research outputs

33 How will participants be informed about the outcome of the study?

N/A

34 How will the results of the study be disseminated and reported?

The results of the study will be reported in a written dissertation submitted as part of the requirements for the MSc Business Analytics degree at the University of Bristol.

35 How will the data that underpins your research findings be made available at the end of the project?

The data that underpins the research findings will be used solely for the purpose of completing the MSc dissertation and will be stored securely for examination purposes. However, it will not be made publicly available, as it includes survey responses and review content that, although anonymised, are not owned.

## Supporting Information

Supporting information Please provide any additional information in relation to your study that you think may be relevant.

Any other information Please upload any other documents that you think may be relevant to your research. There is no limit to the number of documents you can upload.

Documents

| Type | Document Name | File Name | Version Date | Version | Size |
|---|---|---|---|---|---|
| Other | MS forms - Questionnare | MS forms - Questionnare.pdf | | | 155.5 KB |
| Other | Data source, links and terms of use | Data source, links and terms of use.docx | | | 80.0 KB |

**To be completed by your named supervisor**

This free text box below is to be completed by your supervisor.

Your form must now be reviewed by your supervisor confirming that any ethics related issues relating to your proposed research have been addressed and your ethics application is complete.

So that your supervisor can confirm this, please follow these steps:

1. If you have not already done so, share your ethics application with your supervisor by selecting the 'Share' button on the left hand side of the form.
2. Input your supervisor email address in the pop-up box, and grant them 'Read' and 'Write' and 'submit' access.
3. You will then need to email your supervisor separately with the following email requesting them to complete this section:

"Dear [Supervisor Name]

My research ethics application is now ready for you to review and confirm that the ethics related issues relating to my proposed research have been addressed and my ethics application is complete. Can you please click on the link below, complete the free text in the ethics application form to confirm this.

[Insert application URL]

All the best"

Once this section has been completed by your supervisor you please select on the next page, and begin the Ethics Academic Checker declaration process on the next page.

I believe the student has addressed potential ethical concerns and approve the application

**Supervisor Guidance Note:**

Please state your name and the date of signature.

Tian Han, 29/06/2025

To proceed to the next page select 'Next' in the Actions tiles.

To save your application for completion and submission at a later date please select 'Save' in the Actions tiles.

## Business School Ethics Academic Checker Declaration

Guidance Note

In order to ensure that your ethics application is submitted successfully please follow the steps below.

Your form must now be reviewed by your assigned Academic Ethics Checker. They will determine whether your ethics application requires further review. So that your Academic Ethics Checker can complete their declaration the following steps must be completed.

1. Share your ethics application with the Academic Ethics Checker by selecting the 'Share' button on the left hand side of the form.
2. Input your Academic Ethics Checker's email address in the pop-up box, and grant them 'Read' and 'Write' access.
3. You will then need to email your Academic Ethics Checker separately with the following email requesting them to complete this section:

   "Dear [Ethics Academic Checker Name]

   My research ethics application is now ready for you to review and sign prior to submission. Can you please click on the link below, complete the Academic Ethics Checker declaration page and select the signature button.

   [Insert application URL]

   All the best"

Once the form has been signed by your named Academic Ethics Checker you will then need to complete the signature declaration on the next page.

Business School Ethics Checker Determination - Please tick the appropriate box below:

- ☑ I have reviewed the student's proposed research project, and believe that no further ethical review is necessary.
- ☐ I have reviewed the student's proposed research project, and recommend that the application is reviewed by the Business School Research Ethics Committee

Academic Ethics Checker Signature

**Signed:** This form was signed by Dr Xin Fei (qc20250@bristol.ac.uk) on 29/06/2025 18:28

## Signatures

Applicant Signature

I confirm that my responses are complete and accurate.

I understand that any errors or omissions may cause a delay in the processing of my application.

**Signed:** This form was signed by Mr Gokul Kumar (ha24467@bristol.ac.uk) on 30/06/2025 11:12

Supervisor signature request

As the named supervisor please confirm:

1. I have read and reviewed this ethics application.
2. I am satisfied with the content and completeness of this ethics application.

**Signed:** This form was signed by Dr Tian Han (tian.han@bristol.ac.uk) on 30/06/2025 11:21

**Submission Reminder**

**Please not**e - Once all signatures have been obtained. You must ensure you select the **Submit** button on the left hand side of the form.

If you are unsure whether your application has been submitted, please contact business-school-ethics@bristol.ac.uk.

## APPENDIX II - ONLINE QUETIONNAIRE

# How Online Reviews Influence Skincare Product Perception

12 Aug 2025

This short survey (approx. 10 minutes) explores how consumers interpret skincare product reviews; including how review tone, trust, and negativity influence their perceptions and purchase decisions. All responses are anonymous and used only for a university dissertation.

* Required

### Participant Information & Consent

**Brief Project Outline:** This study explores how people interpret and react to online skincare product reviews, focusing on the tone of the reviews (positive or negative) and how trustworthy the reviews appear. The research aims to understand how these factors influence consumers' opinions about skincare reviews and their likelihood of purchasing them. The study is part of my MSc dissertation project at the University of Bristol.

**Why have I been invited to participate?** You have been invited because you are a skincare product user and familiar with online product reviews. Participants are being recruited via convenience and voluntary sampling through university networks and social media. Around 300–500 individuals will participate in the study.

**Do I have to take part?** No. Participation is completely voluntary. You can choose whether or not to take part and can skip any question you're not comfortable with.

**Can I withdraw at any time?** Yes. You can withdraw from the study at any point before submitting the survey, and you do not have to give a reason. However, because responses are anonymous, it will not be possible to identify and remove individual data once the survey has been submitted**.**

**What do I have to do? –** You will be asked to complete a short online survey (approximately 10–15 minutes). The survey includes review samples and questions about how credible or persuasive you find them, and how they might affect your perception of a skincare brand or product. No audio or video recording will be used.

**How will the findings be used? –** The findings will be used for my MSc business analytics dissertation. Results may also be used for academic publication. A summary of findings can be shared with participants upon request.

**What are the possible advantages/disadvantages/risks of taking part? –** There are no direct risks or personal benefits to participating. However, your responses will help improve understanding of consumer behaviour in digital skincare marketing. All questions are opinion-based and non-sensitive

**Will my participation in the study be kept confidential?** – Yes. All responses will be anonymised and stored securely. No names, email addresses, or identifying personal details will be collected. Participants will not be identifiable in any reports, presentations, or publications.

**What will happen to the data collected? –** All data will be stored securely in password-protected files and accessed only by the researcher and supervisor. The data will be anonymised and used strictly for academic purposes. It will be deleted within six months after dissertation submission, in line with University of Bristol data policies.

**Who has reviewed the study?** This Study is being supervised by Professor Tian Han, school of management, business school, University of Bristol. The project has been submitted for full ethical review to the **Business school Research Ethics Committee.**

**Please answer the following questions to the best of your knowledge**

**If you have any concerns related to your participation in this study please direct them to the Faculty of Social Science Research Ethics Committee, via Liam McKervey, Research Governance and Ethics Officer (Tel: 0117 331 7472, email: Liam.McKervey@bristol.ac.uk)**

1. Have you been given information explaining about the study? *

◯ Yes

◯ No

2. Have you received enough information about the study for you to make a decision about your participation? *

◯ Yes

◯ No

3. Do you understand that you are free to withdraw from the study and free to withdraw your data prior to final consent or submission at any time? *

◯ Yes

◯ No

4. Do you understand that you are free to withdraw from the study and free to withdraw your data prior to final consent without having to give a reason for with drawing? *

◯ Yes

◯ No

5. Do you understand that once you submit the survey, your individual data cannot be withdrawn due to anonymous nature of data collection *

◯ Yes

◯ No

6. Do you consent to participating in this research study? *

◯ Yes

◯ No

7. Final Consent? *

Please select 2 options.

☐ I confirm that I am 18 years of age or above

☐ I hereby fully and freely consent to my participation in this study

8. I usually read online reviews before purchasing a skincare product. *

| Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

9. I trust detailed reviews morethan the short ones *

| Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

10. I trust reviews that are marked as "verified Purchase" *

| Strongly Disagree | Disagree | Neutral | Agree | Strongly agree |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

11. What does the "Verified Purchase" tag mean to you? *

○ Reviewer genuinely bought the product

○ Review is more authentic

○ It prevents fake reviews

○ It doesn't affect my judgment

12. A single strongly negative review can make me avoid a product, even if other reviews are positive. *

| Strongly disagree | Disagree | Neutral | Agree | Strongly agree |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

13. **Scenario A** (positive review)

"This moisturiser really improved my skin texture within a week. It's non-greasy and absorbs quickly. I'll definitely buy it again!"

How trustworthy do you find this review? (1- least likely, 5- Most likely) *

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

14. How likely are you to purchase the product based on this review? (1- least likely, 5- Most likely) *

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

15. **Scenario B (Negative Review):**

"This product caused redness and broke me out after two uses. Would not recommend it to anyone with sensitive skin."

How trustworthy do you find this review? (1- least likely, 5- Most likely) *

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

16. How likely are you to avoid purchasing the product based on this review? (1- least likely, 5- Most likely) *

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ |

17. **Scenario C (Short, Unverified):**

"Didn't work for me." *(Unverified)*

**Scenario D (Detailed, Verified):**

"I have combination skin, and this cream helped balance it out. Lightweight, smells nice, and worked well under makeup." *(Verified Purchase)*

**Which review do you find more credible?** *

- ◯ Scenario C (Short, Unverified)
- ◯ Scenario D (Detailed, Verified)
- ◯ Both equally
- ◯ Neither

18. How likely are you to trust reviews that are both verified and detailed?(1- least likely, 5- Most likely) *

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| ◯ | ◯ | ◯ | ◯ | ◯ |

19. **Scenario E (Positive):**

"Great consistency and feel. Works as promised. Will repurchase!"

**Scenario F (Negative):**

"My skin broke out in bumps and stung every time I applied it."
Which review would influence your perception of the product more? *

- ◯ Scenario E (Positive)
- ◯ Scenario F (Negative)
- ◯ Both equally
- ◯ Neither

20. Why did you choose that review? (Select the most relevant reason) *

- ◯ It mentioned a skin reaction or safety issue (e.g., breakouts, irritation)
- ◯ It was emotionally strong or concerning
- ◯ It described actual results or benefits clearly
- ◯ It sounded more authentic or relatable
- ◯ I usually pay more attention to positive reviews
- ◯ I usually pay more attention to negative reviews
- ◯ Other

21. **Scenario G (Short, Unverified):**

"Worst product I've used." *(Unverified)*
**Scenario H (Detailed, Verified):**

"I was really hoping this would help, but it gave me a rash and clogged my pores. Disappointed after seeing good reviews." *(Verified Purchase)*

**Which negative review would influence your decision more?** *

- ◯ Scenario G (Short, Unverified)
- ◯ Scenario H (Detailed, Verified)
- ◯ Both equally
- ◯ Neither

22. Do you believe a credible negative review is more harmful to a brand than a short, vague one? *

| Strongly Disagree | Disagree | Neutral | Agree | Strongly agree |
|---|---|---|---|---|
| ◯ | ◯ | ◯ | ◯ | ◯ |

23. What makes a negative review stand out to you? *

- ◯ Emotional or angry tone
- ◯ Detailed complaints
- ◯ Mention of side effects or skin damage
- ◯ Low star rating
- ◯ Lots of "Not helpful" votes
- ◯ None of the above

24. What is your age group? *

- ◯ 18–24
- ◯ 25–34
- ◯ 35–44
- ◯ 45+

25. How often do you purchase skincare products? *

- ◯ Weekly
- ◯ Monthly
- ◯ Every 2–3 months
- ◯ Rarely

## APPENDIX III DATASETS

### Survey Dataset:

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Id | I usually read | I trust detailed | I trust reviews | What does | A single strong | Scenario | How likely are | Scenario B | How likely | Scenario C | How likely are | Scenario E | Why did you c | Scenario G | Do you believe | What makes a | What is your | How often do y | Q_Avoid_Nega | Q_CredibleNe | Age_num | Purchase_frequency_num | |
| 2 | 2 | 4 | 5 | 4 | Reviewer gen | 2 | 3 | 4 | 3 | 3 | Both equally | 4 | ,Ä†Scenario E | It described a | Scenario H (D | 4 | Low star rating | 45+ | Monthly | 2 | 4 | 47.5 | 2 | |
| 3 | 3 | 5 | 5 | 5 | It prevents fak | 3 | 4 | 3 | 4 | 3 | Scenario D (D | 4 | ,Ä†Scenario F | I usually paym | Scenario H (D | 5 | Detailed com | 18√±24 | Every 2√±3 mo | 3 | 5 | 21 | 3 | |
| 4 | 4 | 5 | 5 | 5 | Review is mor | 2 | 4 | 3 | 3 | 3 | Both equally | 5 | Both equally | It was emotion | Both equally | 4 | Detailed com | 45+ | Monthly | 2 | 4 | 47.5 | 2 | |
| 5 | 5 | 4 | 4 | 4 | It prevents fak | 5 | 3 | 3 | 4 | 5 | Scenario D (D | 3 | Both equally | It described a | Scenario H (D | 5 | Lots of √¨Not | 35√±44 | Every 2√±3 mo | 5 | 5 | 39.5 | 3 | |
| 6 | 7 | 5 | 5 | 5 | Reviewer gen | 4 | 3 | 3 | 4 | 4 | Scenario D (D | 5 | ,Ä†Scenario F | ,Ä†I usually pa | Scenario H (D | 5 | Detailed com | 25√±34 | Monthly | 4 | 5 | 29.5 | 2 | |
| 7 | 9 | 4 | 4 | 4 | Reviewer gen | 4 | 3 | 3 | 4 | 5 | Scenario D (D | 4 | Both equally | It described a | Both equally | 4 | Mention of sic | 18√±24 | Weekly | 4 | 4 | 21 | 1 | |
| 8 | 10 | 4 | 4 | 4 | Review is mor | 2 | 2 | 2 | 4 | 4 | Scenario D (D | 4 | ,Ä†Scenario E | ,Ä†I usually pa | Scenario H (D | 4 | Mention of sic | 45+ | Monthly | 2 | 4 | 47.5 | 2 | |
| 9 | 11 | 5 | 4 | 4 | Review is mor | 3 | 4 | 4 | 5 | 4 | Both equally | 3 | ,Ä†Scenario F | ,Ä†I usually pa | Scenario G (Sl | 4 | Low star rating | 25√±34 | Monthly | 3 | 4 | 29.5 | 2 | |
| 10 | 14 | 3 | 5 | 3 | Reviewer gen | 3 | 4 | 4 | 4 | 4 | Both equally | 5 | Both equally | It was emotion | Both equally | 3 | Mention of sic | 18√±24 | Every 2√±3 mo | 3 | 3 | 21 | 3 | |
| 11 | 16 | 5 | 3 | 4 | Review is mor | 2 | 3 | 3 | 3 | 3 | Scenario D (D | 4 | Both equally | ,Ä†I usually pa | Scenario H (D | 4 | Lots of √¨Not | 35√±44 | Every 2√±3 mo | 2 | 4 | 39.5 | 3 | |
| 12 | 17 | 4 | 4 | 4 | It doesn√≠t af | 3 | 4 | 5 | 5 | 5 | ,Ä†Scenario C | 5 | Both equally | I usually paym | Scenario G (Sl | 4 | Mention of sic | 18√±24 | Rarely | 3 | 4 | 21 | 4 | |
| 13 | 18 | 5 | 5 | 5 | Review is mor | 4 | 5 | 5 | 5 | 5 | Both equally | 5 | Both equally | It described a | Both equally | 4 | Lots of √¨Not | 18√±24 | Monthly | 4 | 4 | 21 | 2 | |
| 14 | 19 | 4 | 5 | 4 | Review is mor | 4 | 3 | 4 | 3 | 3 | Scenario D (D | 3 | ,Ä†Scenario F | It described a | Scenario H (D | 3 | Low star rating | 35√±44 | Monthly | 4 | 3 | 39.5 | 2 | |
| 15 | 20 | 3 | 4 | 4 | Review is mor | 3 | 4 | 4 | 4 | 4 | Scenario D (D | 4 | ,Ä†Scenario F | ,Ä†It mentione | Scenario H (D | 3 | Mention of sic | 45+ | Every 2√±3 mo | 3 | 3 | 47.5 | 3 | |
| 16 | 21 | 4 | 4 | 4 | Review is mor | 3 | 4 | 4 | 4 | 4 | Scenario D (D | 4 | ,Ä†Scenario E | ,Ä†It mentione | Scenario H (D | 4 | Low star rating | 25√±34 | Weekly | 3 | 4 | 29.5 | 1 | |
| 17 | 22 | 3 | 3 | 3 | It doesn√≠t af | 3 | 5 | 4 | 5 | 5 | Both equally | 4 | ,Ä†Scenario E | ,Ä†I usually pa | Both equally | 5 | Mention of sic | 18√±24 | Weekly | 3 | 5 | 21 | 1 | |
| 18 | 23 | 4 | 4 | 3 | Review is mor | 3 | 4 | 3 | 3 | 4 | Scenario D (D | 4 | ,Ä†Scenario F | ,Ä†It sounded | Scenario H (D | 4 | Mention of sic | 25√±34 | Weekly | 3 | 4 | 29.5 | 1 | |
| 19 | 24 | 4 | 5 | 4 | Reviewer gen | 5 | 5 | 4 | 5 | 4 | ,Ä†Scenario C | 4 | ,Ä†Neither | ,Ä†It sounded | Both equally | 4 | Mention of sic | 18√±24 | Weekly | 5 | 4 | 21 | 1 | |
| 20 | 25 | 3 | 4 | 5 | Reviewer gen | 4 | 4 | 4 | 4 | 5 | ,Ä†Scenario C | 4 | ,Ä†Scenario E | ,Ä†It mentione | Scenario G (Sl | 5 | Detailed com | 25√±34 | Monthly | 4 | 5 | 29.5 | 2 | |
| 21 | 26 | 3 | 3 | 4 | It prevents fak | 3 | 3 | 4 | 3 | 4 | ,Ä†Scenario C | 4 | Both equally | It described a | Both equally | 3 | Low star rating | 25√±34 | Weekly | 3 | 3 | 29.5 | 1 | |
| 22 | 27 | 5 | 5 | 5 | Review is mor | 4 | 5 | 5 | 4 | 5 | ,Ä†Scenario C | 5 | ,Ä†Scenario E | It was emotion | Scenario G (Sl | 4 | Mention of sic | 18√±24 | Weekly | 4 | 4 | 21 | 1 | |
| 23 | 29 | 4 | 4 | 4 | Review is mor | 4 | 4 | 4 | 4 | 4 | Both equally | 4 | Both equally | ,Ä†I usually pa | Both equally | 4 | Detailed com | 25√±34 | Monthly | 4 | 4 | 29.5 | 2 | |
| 24 | 30 | 4 | 4 | 4 | Review is mor | 4 | 4 | 4 | 4 | 4 | Both equally | 4 | Both equally | ,Ä†I usually pa | Both equally | 4 | Detailed com | 25√±34 | Monthly | 4 | 4 | 29.5 | 2 | |
| 25 | 31 | 5 | 5 | 5 | It doesn√≠t af | 5 | 5 | 5 | 5 | 5 | Both equally | 5 | Both equally | I usually paym | Both equally | 5 | Low star rating | 18√±24 | Every 2√±3 mo | 5 | 5 | 21 | 3 | |
| 26 | 32 | 5 | 5 | 5 | Reviewer gen | 5 | 5 | 5 | 5 | 5 | Scenario D (D | 5 | Both equally | ,Ä†I usually pa | Scenario G (Sl | 3 | Detailed com | 25√±34 | Rarely | 5 | 3 | 29.5 | 2 | |
| 27 | 33 | 3 | 4 | 3 | It prevents fak | 3 | 3 | 4 | 3 | 3 | Both equally | 4 | ,Ä†Scenario F | ,Ä†I usually pa | Both equally | 4 | Low star rating | 18√±24 | Weekly | 3 | 4 | 21 | 1 | |
| 28 | 34 | 4 | 4 | 4 | It prevents fak | 4 | 4 | 4 | 4 | 4 | Scenario D (D | 4 | Both equally | I usually paym | Both equally | 4 | Detailed com | 18√±24 | Rarely | 4 | 4 | 21 | 4 | |
| 29 | 35 | 3 | 4 | 3 | Reviewer gen | 4 | 4 | 5 | 4 | 5 | Scenario D (D | 4 | ,Ä†Scenario F | I usually paym | Both equally | 4 | Emotional or a | 25√±34 | Rarely | 4 | 4 | 29.5 | 4 | |
| 30 | 36 | 4 | 3 | 3 | Review is mor | 5 | 5 | 5 | 5 | 5 | Both equally | 5 | ,Ä†Scenario F | It was emotion | Scenario H (D | 5 | Lots of √¨Not | 18√±24 | Rarely | 5 | 5 | 21 | 4 | |
| 31 | 37 | 3 | 3 | 4 | It prevents fak | 2 | 4 | 2 | 1 | 3 | Scenario D (D | 3 | Both equally | It described a | Scenario H (D | 4 | Mention of sic | ,Ä†45+ | Rarely | 2 | 4 | 47.5 | 4 | |
| 32 | 38 | 4 | 4 | 5 | Review is mor | 5 | 3 | 4 | 5 | 4 | ,Ä†Scenario C | 5 | ,Ä†Scenario F | ,Ä†It sounded | Scenario H (D | 5 | Low star rating | 18√±24 | Monthly | 5 | 5 | 21 | 2 | |
| 33 | 39 | 3 | 3 | 3 | It doesn√≠t af | 3 | 3 | 4 | 3 | 3 | Both equally | 3 | ,Ä†Scenario F | ,Ä†It sounded | Scenario H (D | 3 | Low star rating | 18√±24 | Monthly | 3 | 3 | 21 | 2 | |
| 34 | 41 | 5 | 5 | 5 | Review is mor | 5 | 4 | 5 | 4 | 3 | Scenario D (D | 5 | ,Ä†Scenario F | It described a | Scenario H (D | 5 | Detailed com | 25√±34 | Monthly | 5 | 5 | 29.5 | 2 | |
| 35 | 43 | 3 | 4 | 4 | It prevents fak | 3 | 1 | 2 | 3 | 3 | Scenario D (D | 4 | Both equally | ,Ä†It sounded | Scenario H (D | 4 | Low star rating | ,Ä†45+ | Every 2√±3 mo | 2 | 4 | 47.5 | 3 | |
| 36 | 44 | 4 | 4 | 4 | Review is mor | 4 | 2 | 4 | 1 | 4 | ,Ä†Scenario C | 3 | ,Ä†Scenario F | It was emotion | Scenario H (D | 4 | Detailed com | ,Ä†45+ | Every 2√±3 mo | 4 | 4 | 47.5 | 3 | |
| 37 | 45 | 5 | 4 | 5 | Review is mor | 4 | 2 | 3 | 4 | 2 | Scenario D (D | 5 | ,Ä†Scenario F | I usually paym | Scenario H (D | 5 | Detailed com | 35√±44 | Monthly | 4 | 5 | 39.5 | 2 | |
| 38 | 46 | 5 | 5 | 5 | Review is mor | 5 | 5 | 5 | 5 | 5 | Both equally | 3 | ,Ä†Scenario F | I usually paym | Scenario H (D | 4 | Mention of sic | 18√±24 | Weekly | 5 | 4 | 21 | 1 | |
| 39 | 47 | 4 | 5 | 3 | It prevents fak | 3 | 4 | 5 | 4 | 5 | ,Ä†Scenario C | 5 | ,Ä†Scenario E | ,Ä†It mentione | Scenario H (D | 4 | Detailed com | 25√±34 | Monthly | 3 | 4 | 29.5 | 2 | |
| 40 | 48 | 5 | 4 | 5 | It prevents fak | 4 | 4 | 4 | 4 | 4 | ,Ä†Scenario C | 4 | Both equally | It was emotion | Scenario G (Sl | 4 | Emotional or a | 25√±34 | Weekly | 4 | 4 | 29.5 | 1 | |
| 41 | 49 | 3 | 4 | 5 | It prevents fak | 3 | 3 | 4 | 5 | 3 | Both equally | 4 | ,Ä†Scenario F | ,Ä†It mentione | Scenario G (Sl | 3 | Emotional or a | 18√±24 | Weekly | 3 | 3 | 21 | 1 | |
| 42 | 50 | 5 | 4 | 5 | It doesn√≠t af | 3 | 3 | 4 | 3 | 3 | ,Ä†Scenario C | 4 | ,Ä†Scenario E | It described a | Scenario H (D | 3 | Low star rating | 18√±24 | Every 2√±3 mo | 3 | 3 | 21 | 3 | |
| 43 | 51 | 5 | 5 | 5 | Reviewer gen | 5 | 5 | 4 | 4 | 4 | Scenario D (D | 5 | ,Ä†Scenario E | It was emotion | Scenario G (Sl | 4 | Low star rating | 25√±34 | Weekly | 5 | 4 | 29.5 | 1 | |
| 44 | 52 | 5 | 4 | 5 | It doesn√≠t af | 3 | 3 | 4 | 3 | 3 | ,Ä†Scenario C | 4 | ,Ä†Scenario E | ,Ä†It mentione | Both equally | 4 | Detailed com | 25√±34 | Weekly | 3 | 4 | 29.5 | 1 | |
| 45 | 53 | 4 | 4 | 3 | Reviewer gen | 3 | 3 | 3 | 4 | 4 | Scenario D (D | 4 | ,Ä†Scenario F | ,Ä†It mentione | Both equally | 4 | Mention of sic | 35√±44 | Monthly | 3 | 4 | 39.5 | 2 | |
| 46 | 56 | 4 | 4 | 5 | Review is mor | 4 | 3 | 4 | 3 | 4 | Scenario D (D | 3 | ,Ä†Scenario F | It described a | Scenario H (D | 3 | Mention of sic | 35√±44 | Every 2√±3 mo | 4 | 3 | 39.5 | 3 | |
| 47 | 57 | 4 | 4 | 4 | Review is mor | 4 | 4 | 4 | 2 | 3 | Both equally | 4 | ,Ä†Scenario E | ,Ä†It sounded | Scenario H (D | 3 | Detailed com | 18√±24 | Every 2√±3 mo | 4 | 3 | 21 | 3 | |
| 48 | 58 | 5 | 5 | 5 | Review is mor | 5 | 5 | 5 | 3 | 2 | Scenario D (D | 5 | ,Ä†Scenario F | It was emotion | Scenario H (D | 4 | Detailed com | 25√±34 | Monthly | 5 | 4 | 29.5 | 2 | |
| 49 | 60 | 2 | 3 | 3 | It prevents fak | 2 | 2 | 2 | 3 | 3 | Both equally | 4 | ,Ä†Neither | Just want to ch | Scenario H (D | 4 | None of the ab | 18√±24 | Every 2√±3 mo | 2 | 4 | 21 | 3 | |
| 50 | 61 | 4 | 4 | 4 | Reviewer gen | 3 | 4 | 4 | 3 | 3 | Scenario D (D | 3 | ,Ä†Scenario E | It described a | Scenario H (D | 3 | Mention of sic | 35√±44 | Every 2√±3 mo | 3 | 3 | 39.5 | 3 | |
| 51 | 62 | 4 | 4 | 3 | It prevents fak | 3 | 3 | 3 | 3 | 3 | Both equally | 4 | Both equally | I most read all | Both equally | 4 | Detailed com | 18√±24 | Rarely | 3 | 4 | 21 | 4 | |
| 52 | 64 | 5 | 4 | 3 | It doesn√≠t af | 3 | 3 | 3 | 4 | 4 | Both equally | 5 | ,Ä†Scenario F | It was emotion | Scenario H (D | 4 | Detailed com | 18√±24 | Weekly | 3 | 4 | 21 | 1 | |

**Amazon review Dataset:**

| | A | B | C | D | E | F | G | H | I | J | K | L |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | rating | title | text | asin | parent_asin | user_id | timestamp | helpful_vote | verified_purc | product_title | product_des | combined_text |
| 2 | 5 | Handy | Great for ma | B099DRHW5 | B099DRHW5 | AHREXOGQF | 1.63E+12 | 0 | TRUE | Disposable F | [] | disposable facial cotton tissue, 100pcs cotton towels face cleansing wipes for sensitive skin, makeup remover washcloth dry and wet use [ ] |
| 3 | 5 | Great for at h | This is perfec | B08P2DZB4 | B08P2DZB4 | AFSKPY37N3 | 1.63E+12 | 0 | FALSE | NIRA Skincar | [] | nira skincare laser & serum bundle - includes anti-aging laser & hyaluronic acid serum - reduces appearance of fine lines & wrinkles - fda cleared [ ] |
| 4 | 3 | Not what I th | I was very dis | B08DHTJ25J | B08DHTJ25J | AFSKPY37N3 | 1.63E+12 | 0 | FALSE | Orange Peel | [] | orange peel nature's cleanse facial scrub - by visage pure - usda organic - physician formulated - research supported - natural exfoliating, refreshing, brightening, skin tightening vitamin c scrub. cleans the pores deeply a |
| 5 | 5 | A little goes a | This is a reall | B07RBSLNF | B07RBSLNF | AFSKPY37N3 | 1.62E+12 | 0 | FALSE | OGANA CELL | [] | ogana cell peptide concentrating amazing lotion 2.03 fl.oz. (60ml) - 650ppm peptide contained mild moisturizing facial lotion, hypoallergenic natural ingredients for sensitive skin [ ] |
| 6 | 5 | Great refresh | I had never tr | B08GLG6W8 | B08GLG6W8 | AFSKPY37N3 | 1.61E+12 | 0 | FALSE | Hanhoo Red | [] | hanhoo red pomegranate 6pcs skin care set (6pcs/set) [ ] |
| 7 | 3 | Didn't do mu | When I saw t | B08M3C6LV | B08M3C6LV | AFSKPY37N3 | 1.61E+12 | 0 | FALSE | 24K Gold Un | [] | 24k gold under eye mask‚Äì 20 pairs- under eye patches with face massager roller, anti-aging hyaluronic acid collagen eye masks for dark circles and puffiness dark under eyes treatment [ ] |
| 8 | 5 | Wonderful o | To be honest | B07W397Q | B07W397Q | AFSKPY37N3 | 1.59E+12 | 1 | FALSE | Iryasa Night I | [] | iryasa night indulge cream - natural face cream for dry skin - vegan anti aging night cream for women - firming cream for face and neck - organic vitamin c moisturizer for face - 1.7oz [ ] |
| 9 | 5 | This really wo | At first when | B07GDQPG1 | B07GDQPG1 | AFSKPY37N3 | 1.55E+12 | 0 | FALSE | Spa Grade Ja | [] | spa grade jade roller for face with gua sha \| jade face roller massager \| jade facial roller \| face roller massager \| jade rollers \| wrinkle roller [ ] |
| 10 | 5 | Great combo | I love this co | B01M7UMAL | B01M7UMAL | AFSKPY37N3 | 1.51E+12 | 0 | FALSE | Philips Sonic | ['Improve you | philips sonicare essence+ gum health & airfloss rechargeable electric flosser, bundle value pack, hx8218/02 [ 'improve your oral health with the new philips sonicare essence+ gum h |
| 11 | 1 | not what it c | nothing spec | B01AKTGHF | B01AKTGHF | AHGAOIZVOI | 1.5E+12 | 0 | TRUE | Charcoal Kor | [] | charcoal konjac face sponge 3 pk \| acne, psoriasis, bumpy skin & ingrown hairs [ ] |
| 12 | 2 | Ineffective | A total waste | B083BDVS3 | B083BDVS3 | AHZ6XMOLE | 1.62E+12 | 0 | TRUE | Adofect 31 P | [] | adofect 31 pairs gold eye mask collagen eye gel pads under eye mask for puffiness and dark circle under eye patches for women and men, gold [ ] |
| 13 | 3 | It does work | Be careful th | B07NSR3CK | B07NSR3CK | AH4O5W3EN | 1.56E+12 | 0 | TRUE | Foot Peel Ma | [] | foot peel mask exfoliating (3 pairs) - foot peeling mask (2 pairs) & moisturizing foot mask (1 pairs), make your feet baby soft, peel away calluses and dead skin for women & men [ ] |
| 14 | 5 | Great for sen | I'm loving this | B018F28B1Y | B018F28B1Y | AH4O5W3EN | 1.48E+12 | 2 | TRUE | Merit Advanc | [] | merit advanced eye serum [ ] |
| 15 | 2 | Not sure wha | This is rough | B09CQ4PXL | B09CQ4PXL | AHATA6X6MY | 1.64E+12 | 0 | FALSE | Sonic Facial | [] | sonic facial cleansing brush, electric silicone face brush and massager, waterproof silicone face scrubber for deep cleansing, exfoliating, blackhead removing, rechargeable,for girls gifts [ ] |
| 16 | 4 | Item does wl | This mask su | B08DXDLR3 | B08DXDLR3 | AHATA6X6MY | 1.6E+12 | 0 | FALSE | 3D Mask Bra | [] | 3d mask bracket - internal support holder frame nose breathing smoothly - protect lipstick lips - diy face mask accessories - 5pcs (l-adult) [ ] |
| 17 | 4 | Deep cleanir | 4 stars becau | B07FX94GY | B07FX94GY | AHATA6X6MY | 1.54E+12 | 0 | FALSE | Pore Strips, E | ['Pore strips a | pore strips, blackhead remover charcoal nose strips deep cleansing 24 strips [ 'pore strips are tested under strict us regulations. no fillers, harmful fragrances, dyes or pre |
| 18 | 4 | Hydrating! | This lip butte | B0B8DCC53 | B0B8DZ7H5 | AHV6QCNBJ | 1.66E+12 | 0 | FALSE | Frankies Biki | [] | frankies bikinis lip butter gloss, sleeping vegan lip mask overnight, moisturizing lip plumper, vitamin e lip balm moisturizer, hydrating lip oil with cocoa butter, made in usa, guava 0.7 ounces [ ] |
| 19 | 5 | Luxurious! | I am enjoying | B0B4JP5YD9 | B0B4JP5YD9 | AHV6QCNBJ | 1.66E+12 | 0 | FALSE | ELLI K ESSEN | [] | elli k essential sincerity from az time reverse cream - made in usa - anti-aging face moisturizer for dry & rough skin ‚Äì repairing treatment cream ‚Äì highly concentrated formula, 1.76 oz. [ ] |
| 20 | 4 | Good Way to | This is a good | B0B4JPGX8F | B0B4JPGX8F | AHV6QCNBJ | 1.66E+12 | 0 | FALSE | Elli K Essenti | [] | elli k essential sincerity from az time reverse double ampoule ‚Äì made in usa - double layering serum ‚Äì repairing & deep hydrating ‚Äì oil and cream formula for healthy and smooth skin, (2.02 fl.oz.) [ ] |
| 21 | 4 | Small Bottle | I'm in my mic | B09KT4RJG6 | B09KT4RJG6 | AHV6QCNBJ | 1.64E+12 | 0 | FALSE | Simply Dana | [] | simply dana rivitastore neck firming cream ‚Äì skin firming cream for tightening neck, jawline, and chest, collagen peptides, and squalane cream for wrinkles, sagging, and dryness [ ] |
| 22 | 4 | Creamy! | This night cre | B07KQ32Z8 | B07KQ32Z8 | AHV6QCNBJ | 1.63E+12 | 0 | FALSE | APIVITA Quee | [] | apivita queen bee holistic age defense night cream 1.69 fl.oz. \| intensive night treatment that speeds skin regeneration, smooths wrinkle & increases skin elasticity with royal jelly & hyaluronic acid [ ] |
| 23 | 4 | Smells Good | I enjoy sugar | B07W6H8CC | B07W6H8CC | AHV6QCNBJ | 1.62E+12 | 0 | FALSE | Little Moon E | [] | little moon essentials tropical bath & shower sugar exfoliant, beach all you want, 2 oz. [ ] |
| 24 | 4 | Non-Irritating | I am in my ea | B08KYLTK5H | B08KYLTK5H | AHV6QCNBJ | 1.61E+12 | 0 | FALSE | MORe di Cro | [] | more di croatia eye cream - dubrovnik gold anti aging eye cream for wrinkles and fine lines, reduce puffiness and dark circles, hyaluronic acid and immortelle 15ml [ ] |
| 25 | 4 | Refreshing! | I like wearing | B08KWN77L | B08KWN77L | AHV6QCNBJ | 1.61E+12 | 0 | FALSE | GENSKIN Ger | [] | genskin generation skin hydrating marine algae eye patches \| hydrate, plump, brighten eye area 180g/6.34 oz - 30 pairs [ ] |
| 26 | 5 | Best Face W | I have tried s | B08FRQGYD | B08FRQGYD | AHV6QCNBJ | 1.61E+12 | 0 | FALSE | SeroVital Sup | [] | serovital super saturated restoring cleansing cloths - cleansing face wipes with green tea and hyaluronic acid - makeup remover wipes - 1 pack 15 ct [ ] |
| 27 | 4 | A Bit Oily | I am in my ea | B08BZ1RHP | B08BZ1RHP | AHV6QCNBJ | 1.6E+12 | 3 | FALSE | MD Complet | [] | md complete bright & healthy vitamin c+ vitalizing face serum \| with vitamin c, vitamin e and herbal extracts \| potent antioxidant protection, nourishes, brightens and evens skin 0.5 fl oz [ ] |
| 28 | 4 | Comfortable | I work at hon | B088FBNQX | B088FBNQX | AHV6QCNBJ | 1.59E+12 | 1 | FALSE | MainBasics H | [] | mainbasics headband with buttons for face masks (gray) [ ] |
| 29 | 4 | Leaves Skin I | I am in my ea | B07PRDZ2B | B07PRDZ2B | AHV6QCNBJ | 1.59E+12 | 0 | FALSE | LUA SKINCAF | ['Body Care'] | lua skincare clarity calming face oil, 1 fz [ 'body care' ] |
| 30 | 4 | Light Weight | This is a light | B083TLNBJJ | B083TLNBJJ | AHV6QCNBJ | 1.59E+12 | 0 | FALSE | BE PLAIN Vit | [] | be plain vitamin ampoule 1.01 fl oz. - korean multi vitamin serum for face with vitamin b, vitamin c, vitamin e, hyaluronic acid [ ] |
| 31 | 3 | Not The Righ | I guess I'm nc | B084D86YL8 | B084D86YL8 | AHV6QCNBJ | 1.58E+12 | 0 | FALSE | MD Complet | [] | md complete bright & healthy citrus-c refresh retinol + vitamin c multitasking treatment [ ] |
| 32 | 4 | Nice Fruity S | This serum is | B07WNBZQ | B07WNBZQ | AHV6QCNBJ | 1.58E+12 | 0 | FALSE | SERUM By M | [] | serum by merlot - resveratrol - natural anti-aging - damage corrector - boost collagen - 1 fl oz. - dark circle corrector - [ ] |
| 33 | 4 | Non-Greasy | I apply body l | B07SW7D6Z | B07SW7D6Z | AHV6QCNBJ | 1.58E+12 | 0 | FALSE | Kale Firm & S | [] | kale firm & smooth body lotion - superfood infused firming body lotion with kale. 6.8 fl.oz, anti-aging moisturizing lotion for all skin types [ ] |
| 34 | 4 | Easy to Use! | As I have gott | B082MTTFZD | B082MTTFZD | AHV6QCNBJ | 1.58E+12 | 1 | FALSE | Wow the Cro | [] | wow the crowd eyelash growth serum for full, luscious, long and thick lashes - natural formula (7ml), clinically and dermatologist tested - boost and enhance eyelashes and brows [ ] |
| 35 | 5 | Gentle! | This goat mill | B07NPWK16 | B07NPWK16 | AHV6QCNBJ | 1.58E+12 | 0 | FALSE | 13.5 fl. Oz. G | [] | 13.5 fl. oz. goat milk facial cleanser, moisturizing face wash for women, hydrating natural face wash, anti-aging face wash, face wash for aging ski [ ] |
| 36 | 3 | Doesn't Irrita | I am in my ea | B077YR333 | B077YR333 | AHV6QCNBJ | 1.55E+12 | 0 | FALSE | Anti-Aging Hy | ['Anti-Aging H | anti-aging hydrating serum with natural mango micro-beads, vitamins e and c, natural and organic plant extracts, paraben-free, 30 ml, made in france, phytema [ 'anti-aging hydrating serum with natu |
| 37 | 5 | Great Quality | I have been u | B07GDQPG1 | B07GDQPG1 | AHV6QCNBJ | 1.55E+12 | 0 | FALSE | Spa Grade Ja | [] | spa grade jade roller for face with gua sha \| jade face roller massager \| jade facial roller \| face roller massager \| jade rollers \| wrinkle roller [ ] |
| 38 | 5 | Impressive! | I have one of | B071R2QPF | B071R2QPF | AHV6QCNBJ | 1.54E+12 | 0 | FALSE | AuroTrends M | [] | aurotrends makeup remover towel,2 pack reusable chemical-free microfiber makeup remover cloths,magically and gently wipe away cosmetics,sunscreen,oil and dirt,set of 2 (pink+black) [ ] |
| 39 | 5 | Great Exfolia | I love coffee | B01B6V11U | B01B6V11U | AHV6QCNBJ | 1.48E+12 | 0 | TRUE | Scrub Angel ( | [] | scrub angel organic arabica coffee body scrub 8oz. an exfoliator that leaves skin smooth and soft while reducing cellulite, 100% natural treatment [ ] |
| 40 | 5 | Doesn't Irrita | A few month | B00GUTPV4 | B00GUTPV4 | AHV6QCNBJ | 1.48E+12 | 0 | TRUE | Johnson & Jo | ['Neutrogena | johnson & johnson ltd neutrogena visibly clear blackhead eliminating 7 day rescue [ 'neutrogena visibly clear blackhead eliminating 7 day recsue triple action cleanser 100ml |
| 41 | 5 | Doesn't Sting | I am in my la | B01CO73OI | B01CO73OI | AHV6QCNBJ | 1.47E+12 | 8 | TRUE | Anti Wrinkle I | [] | anti wrinkle eye cream by vebelle anti aging company - top rated under eye, crows feet moisturizer treatment, natural organic instant moisturizing & hydrating, best for fine lines & wrinkles - 1.7oz [ ] |
| 42 | 4 | Best Peeling | This is the 3r | B00KR4AFJU | B00KR4AFJU | AHV6QCNBJ | 1.46E+12 | 0 | FALSE | Silky Smooth | ['Exfoliate Yo | silky smooth foot peeling exfoliating mask - 1 pair - natural foot care treatment - callus remover - removes dead skin from your feet - heel peel care - your feet will feel silky smooth - be confident in your feet - love your fe |
| 43 | 4 | Fun to Use! | My daughter | B015ZXMSF | B015ZXMSF | AHV6QCNBJ | 1.46E+12 | 3 | FALSE | Anti Aging Fa | ['Love going t | anti aging facial masks by erh with intensive care serum to help reduce the appearance of fine lines and wrinkles. new version exclusive to amazon with expanded mask coverage. 5-mask pack [ 'love going to th |
| 44 | 4 | Slight Fading | I burned my l | B00KXFD75 | B00KXFD75 | AHV6QCNBJ | 1.45E+12 | 0 | FALSE | Protege FLAV | [] | protege flawless scar gel cream, best for scar removal, flattens and softens scars (1 ounce) [ ] |
| 45 | 5 | Looks Great! | I am very hap | B010B0S67 | B010B0S67 | AHV6QCNBJ | 1.44E+12 | 1 | FALSE | Blackhead M | [] | blackhead mask,sky-shop facial mask nose mask face mask blackhead cream with activated charcoal deep pore cleanse, peel off deep skin clean purifying face mask for acne and blackheads (100g) (110g) [ ] |
| 46 | 5 | Works very w | Works very w | B08SWV1X6 | B08SWV1X6 | AGYBOBPV4 | 1.65E+12 | 0 | TRUE | Foot Scrubb | [] | foot scrubber files for callus remover - 32 rows easkep professional stainless steel foot file for pedicure to removes hard skin corns for dry and wet feet [ ] |
| 47 | 5 | Adorable and | The headban | B089K4SV23 | B089K4SV23 | AEWTKEIDMS | 1.59E+12 | 0 | TRUE | Headband fo | [] | headband for women washing face and spa headbands by gnawrishing, use for makeup headband, facial headband and hair bands for washing face [ ] |
| 48 | 5 | love it! | I use this eve | B010B5ZE4 | B010B5ZE4 | AGCCFLGNM | 1.49E+12 | 0 | TRUE | Maquita Wat | [] | maquita waterproof portable face facial highlighter stick shimmer powder makeup silver [ ] |
| 49 | 5 | Seems good | Seems good | B09SGXNB2 | B0BP2Y6QY | AFJBKPK5W5 | 1.67E+12 | 0 | FALSE | LUMOSMAX | [] | lumosmax hair growth serum with built-in scalp massager - promotes hair care and improves scalp health while nourishing scalp \| serum applicator for women [ ] |
| 50 | 5 | Works great | Great soap w | B08KHRF9N | B08KHRF9N | AFJBKPK5W5 | 1.62E+12 | 0 | FALSE | Calendula Sc | [] | calendula soap with coconut oil soap - natural body soap \| best body bar for dry & sensitive skin \| use as a bath soap & face pure extracts (2.64 ounce) [ ] |

**Code Google collab:**

https://colab.research.google.com/drive/1FBJgK83H_9juQ7r-3CDWUVZ33iSv4xTN?usp=sharing